

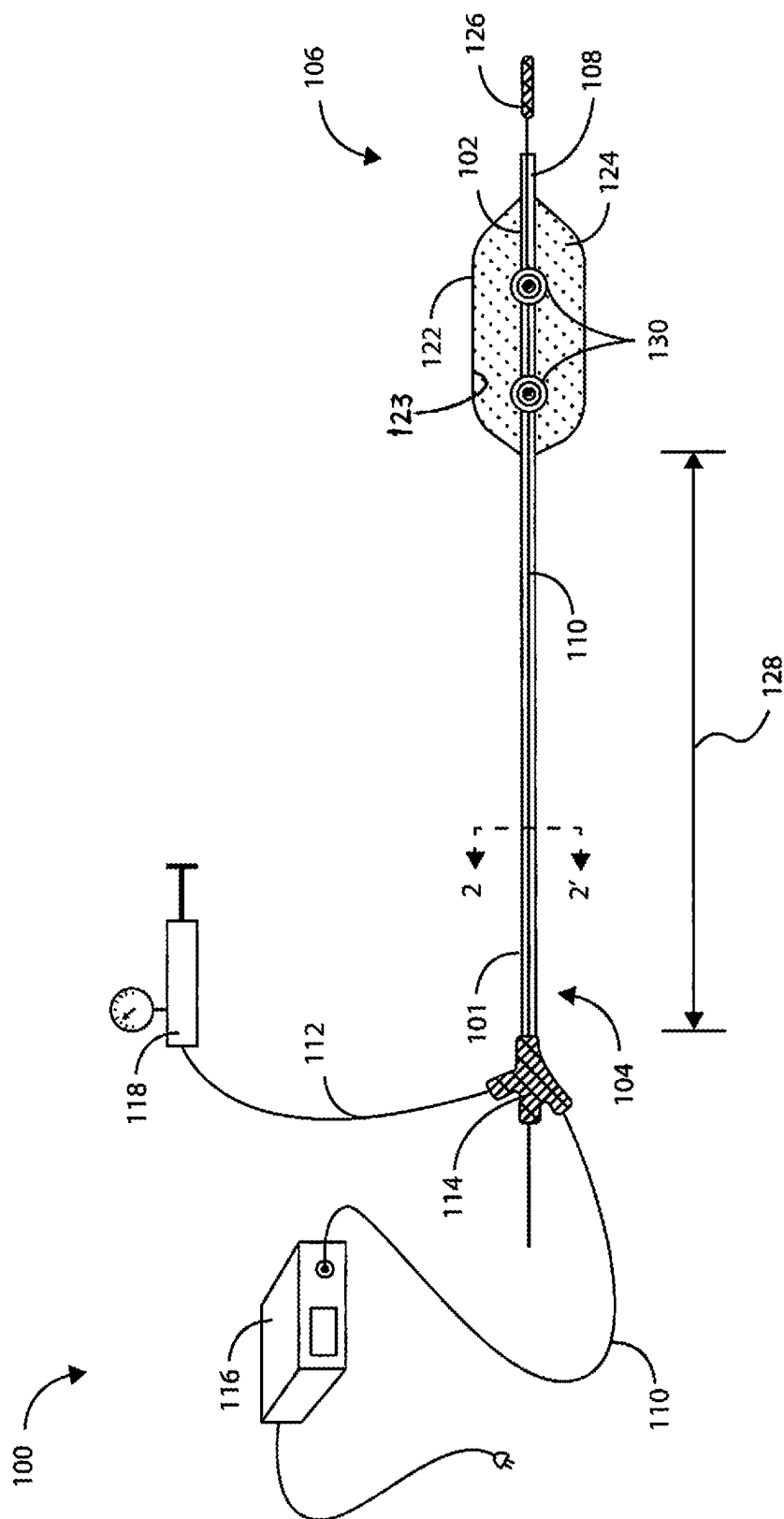


FIG. 1

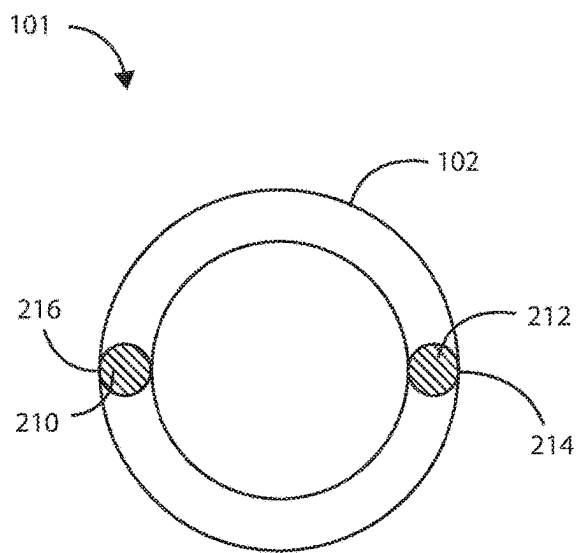


FIG. 2

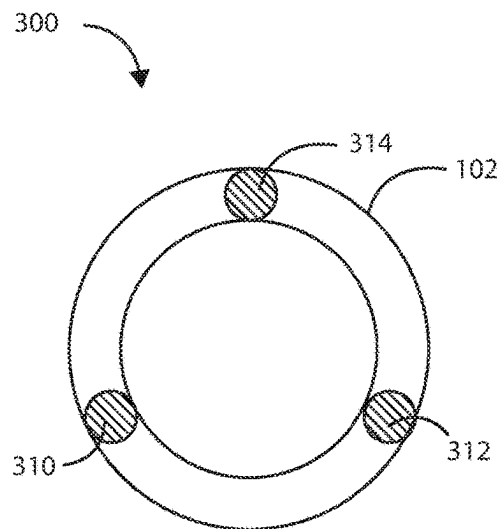


FIG. 3

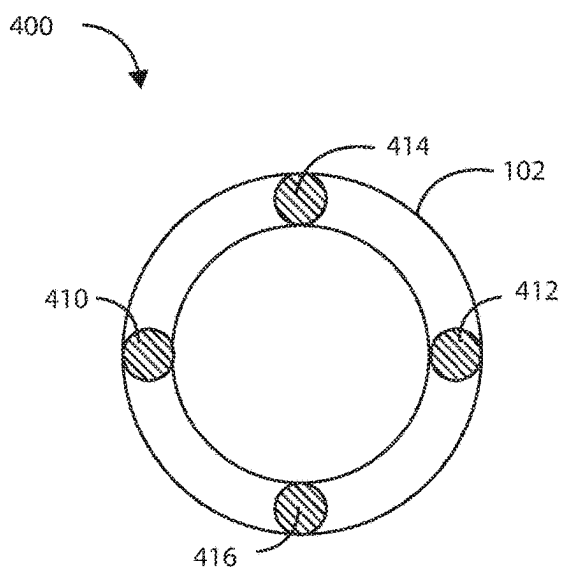


FIG. 4

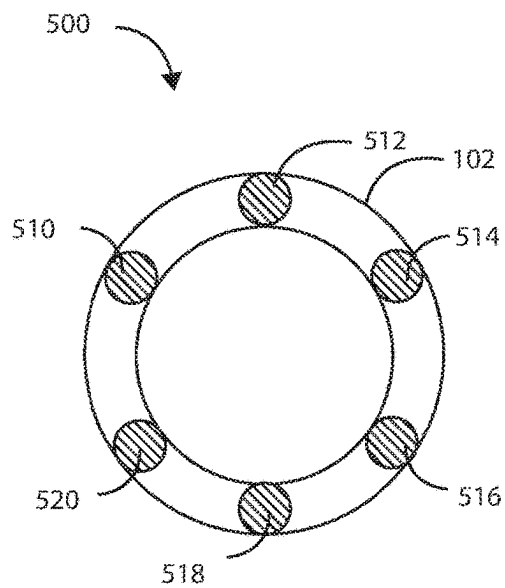


FIG. 5

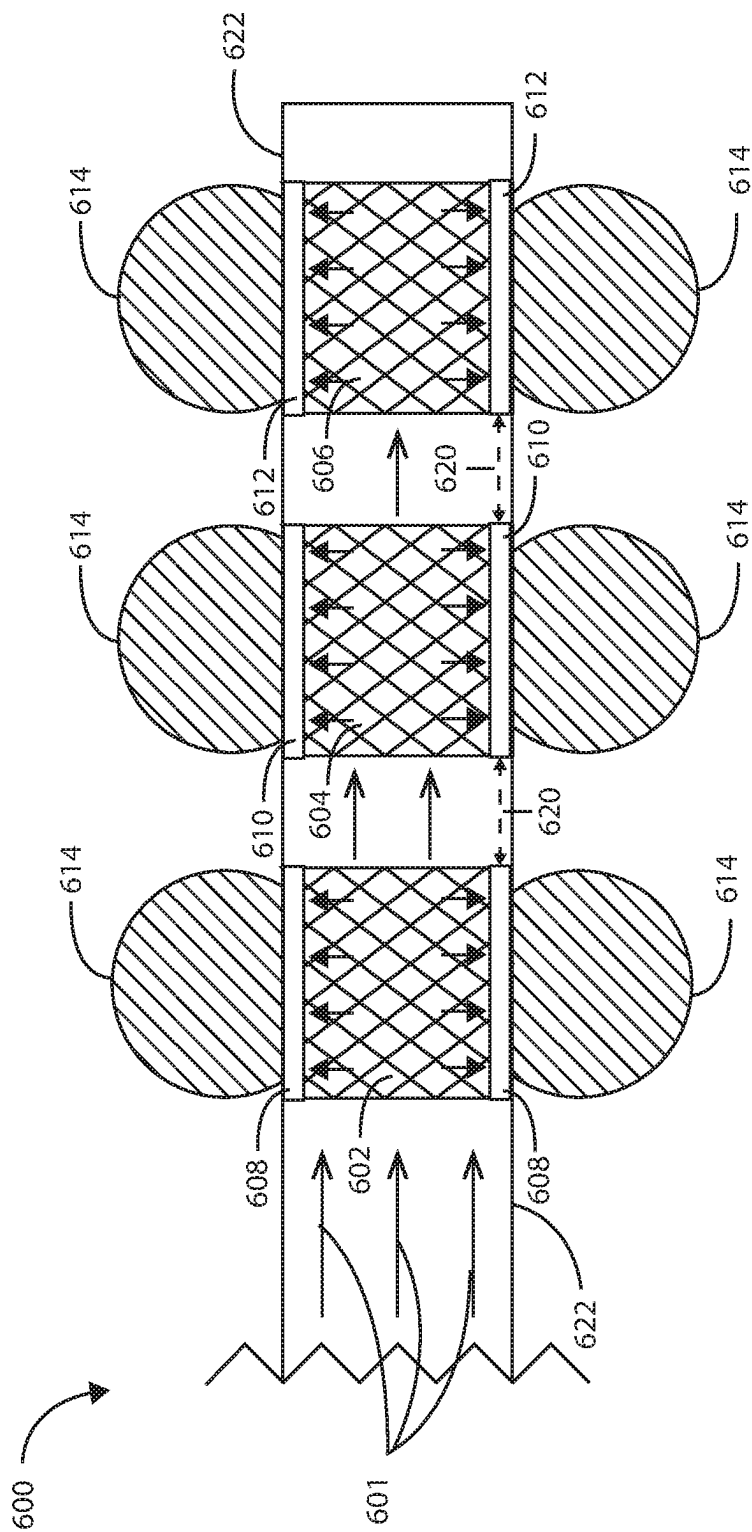


FIG. 6

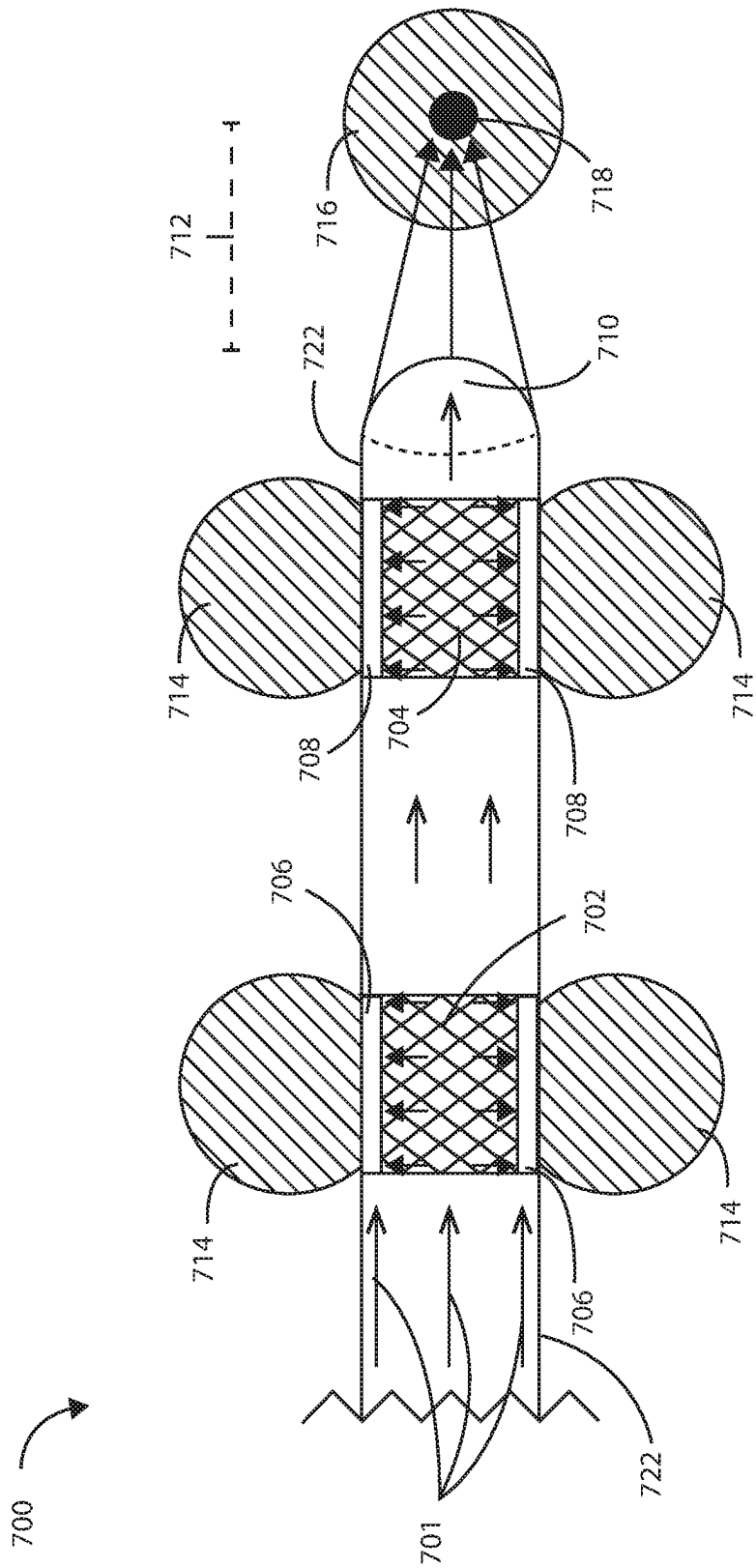


FIG. 7

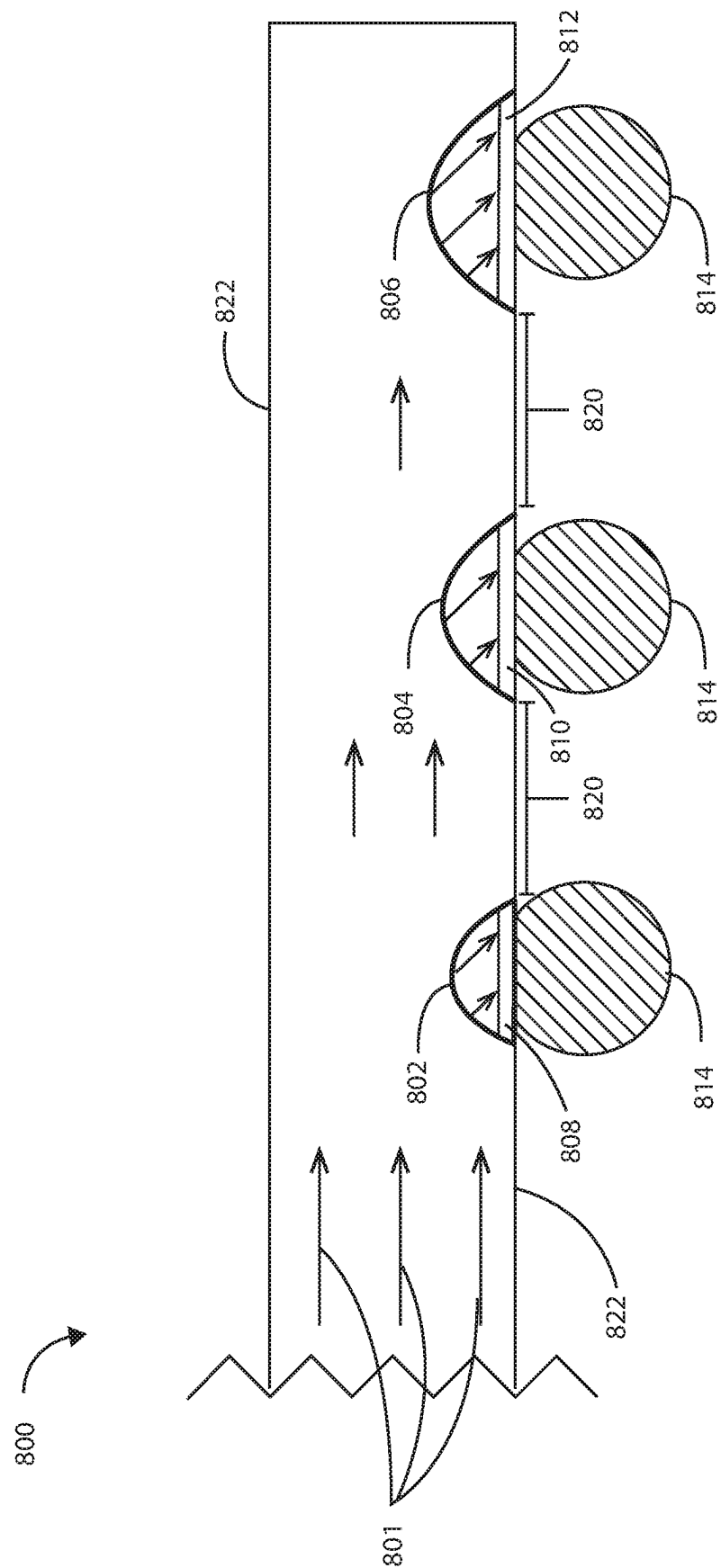
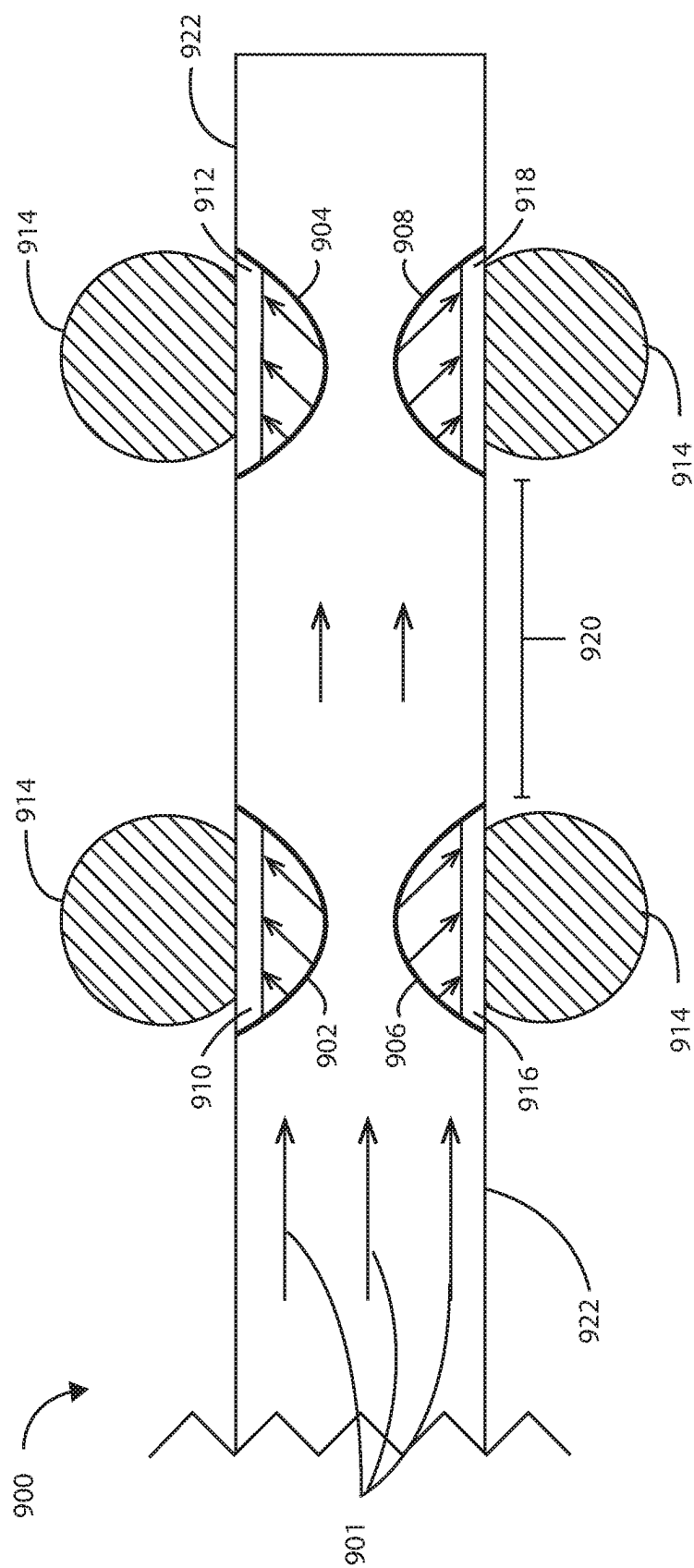


FIG. 8



9
6^{*}
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L

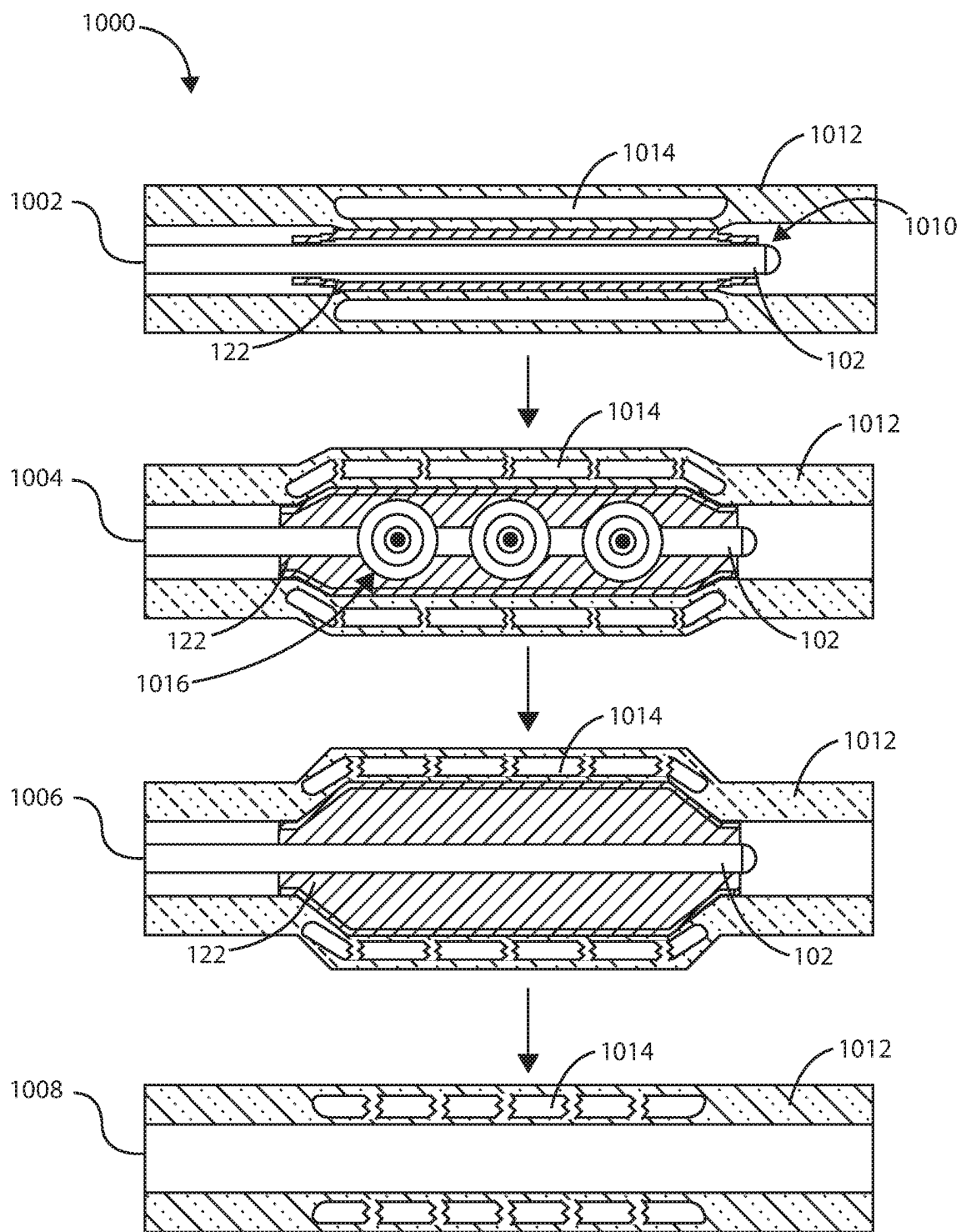


FIG. 10

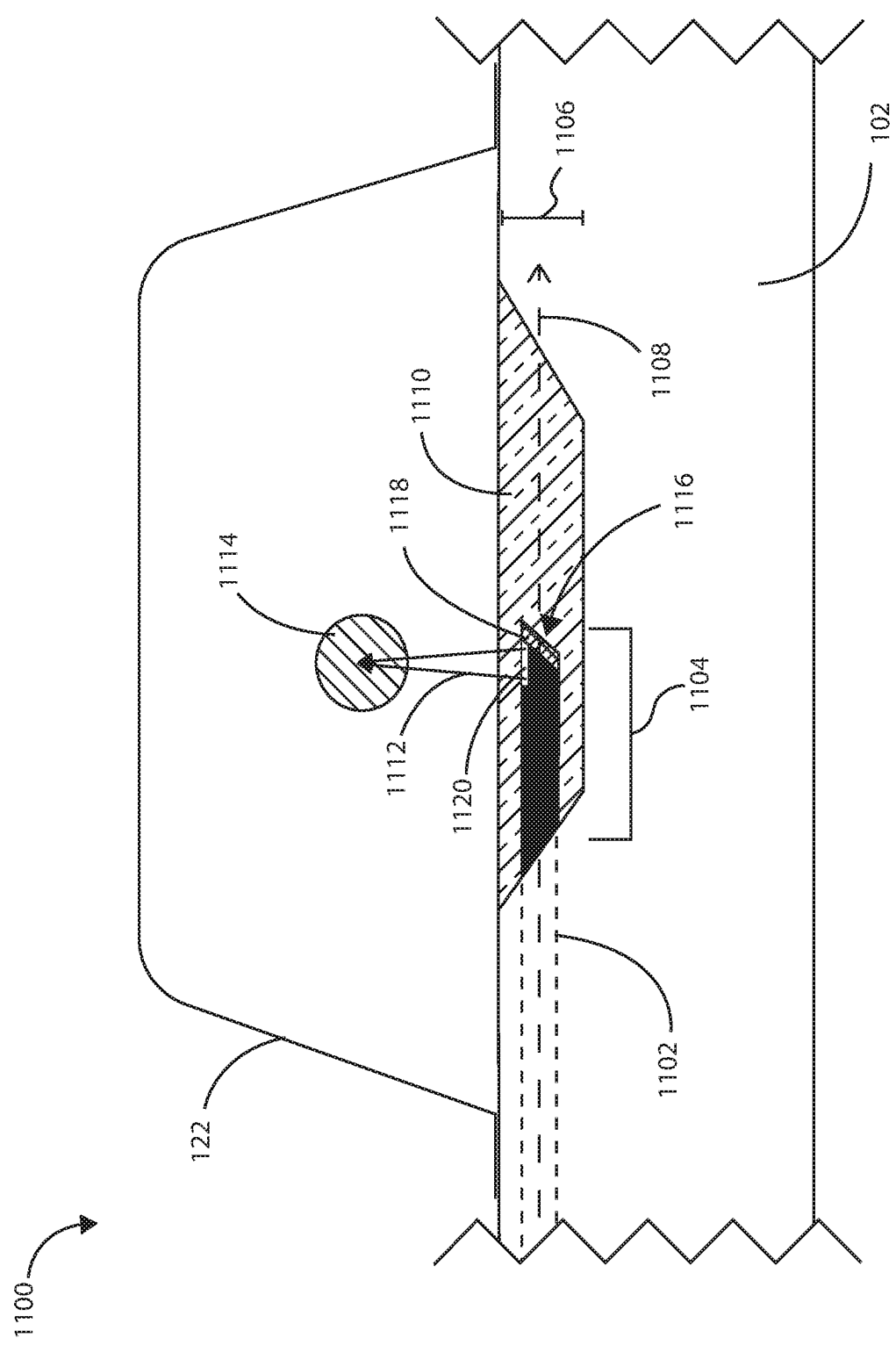


FIG. 11

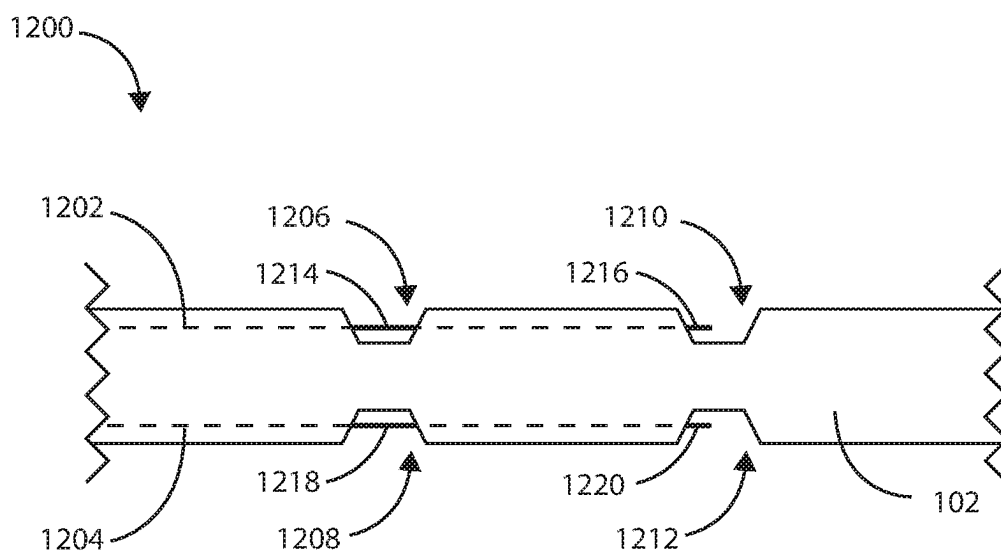


FIG. 12

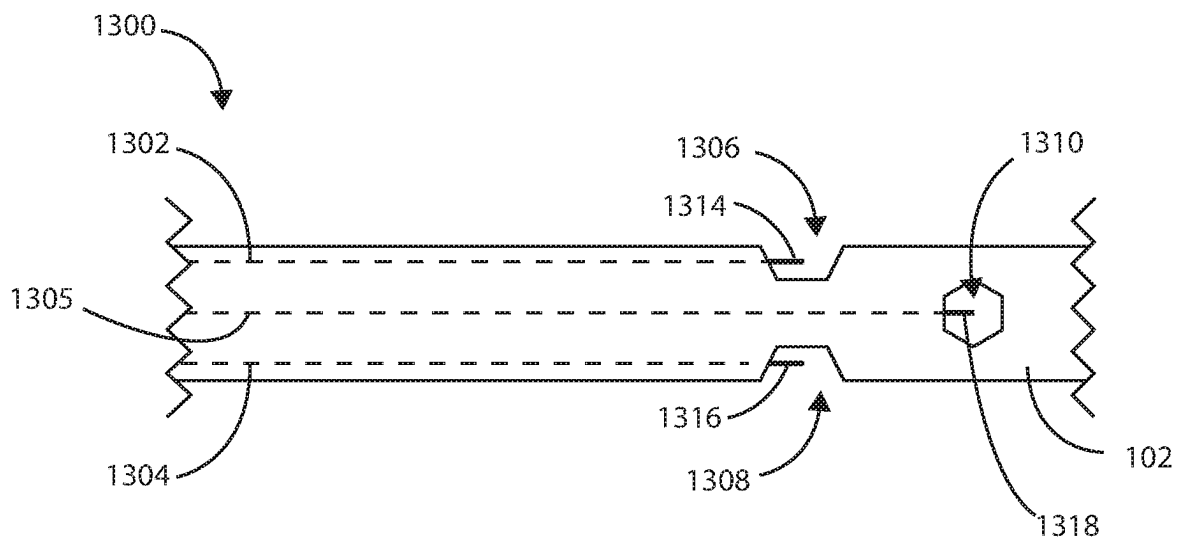


FIG. 13

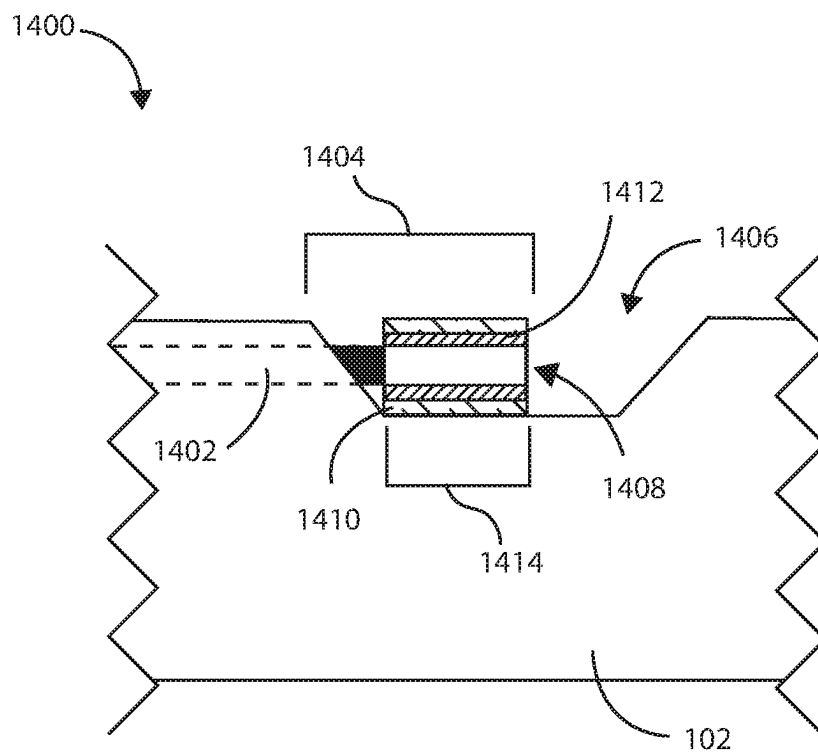


FIG. 14

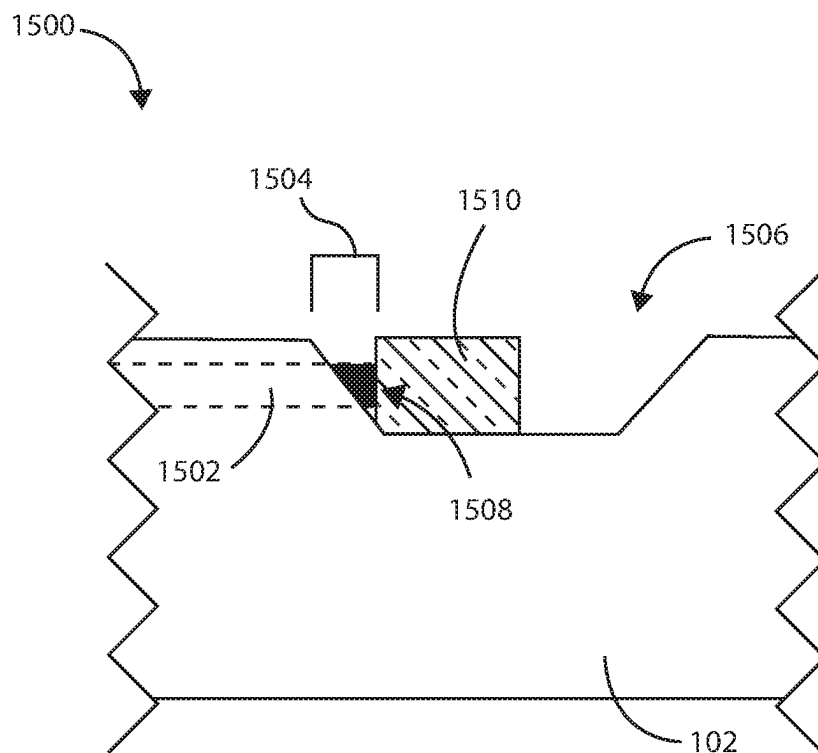


FIG. 15

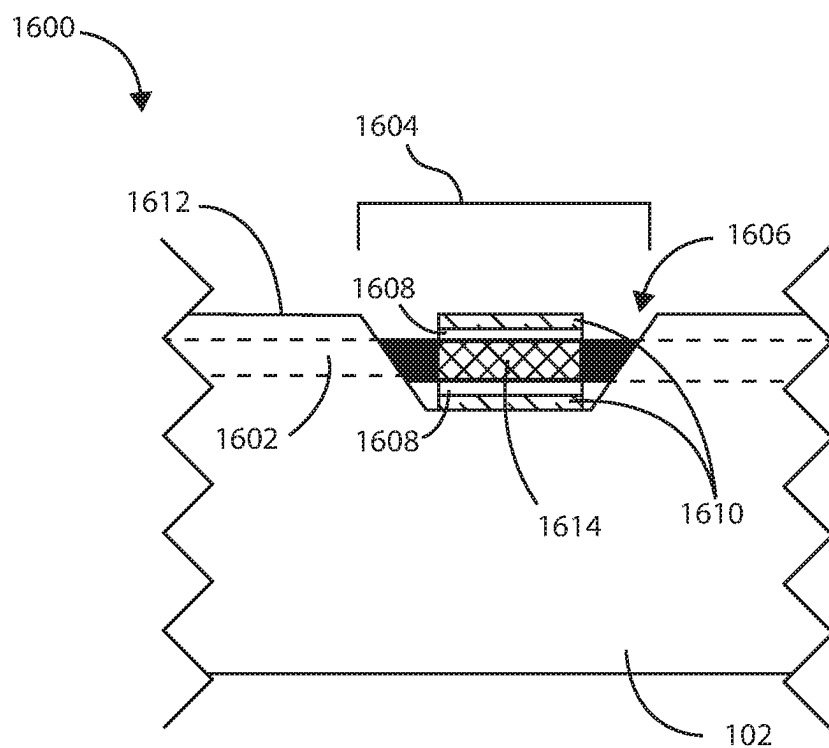


FIG. 16

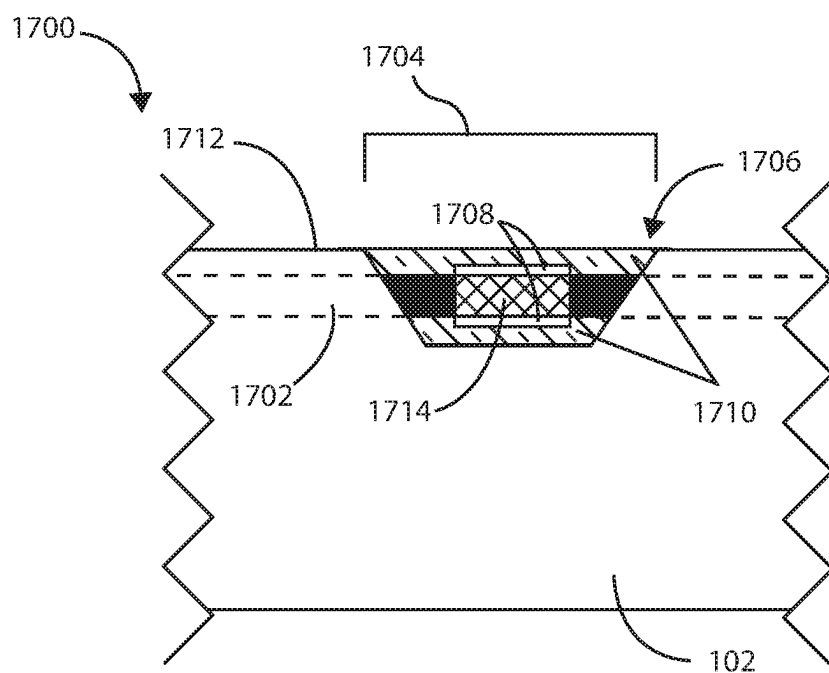


FIG. 17

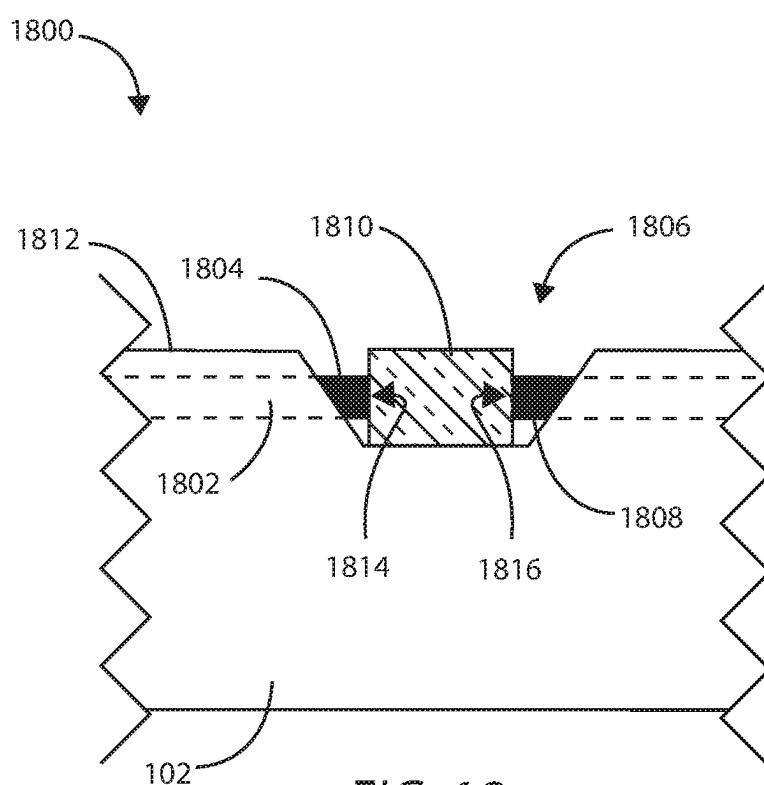


FIG. 18

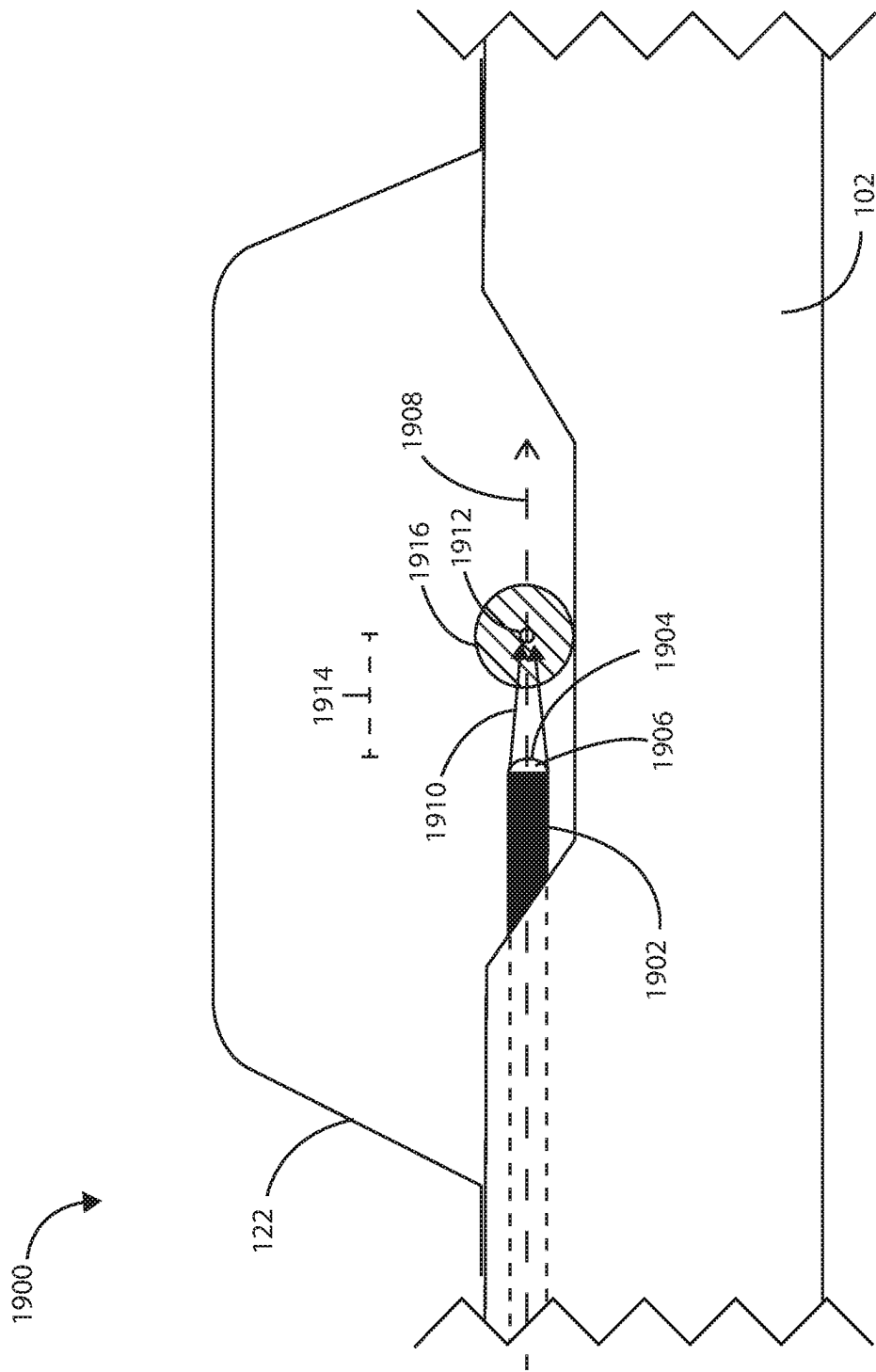


FIG. 19

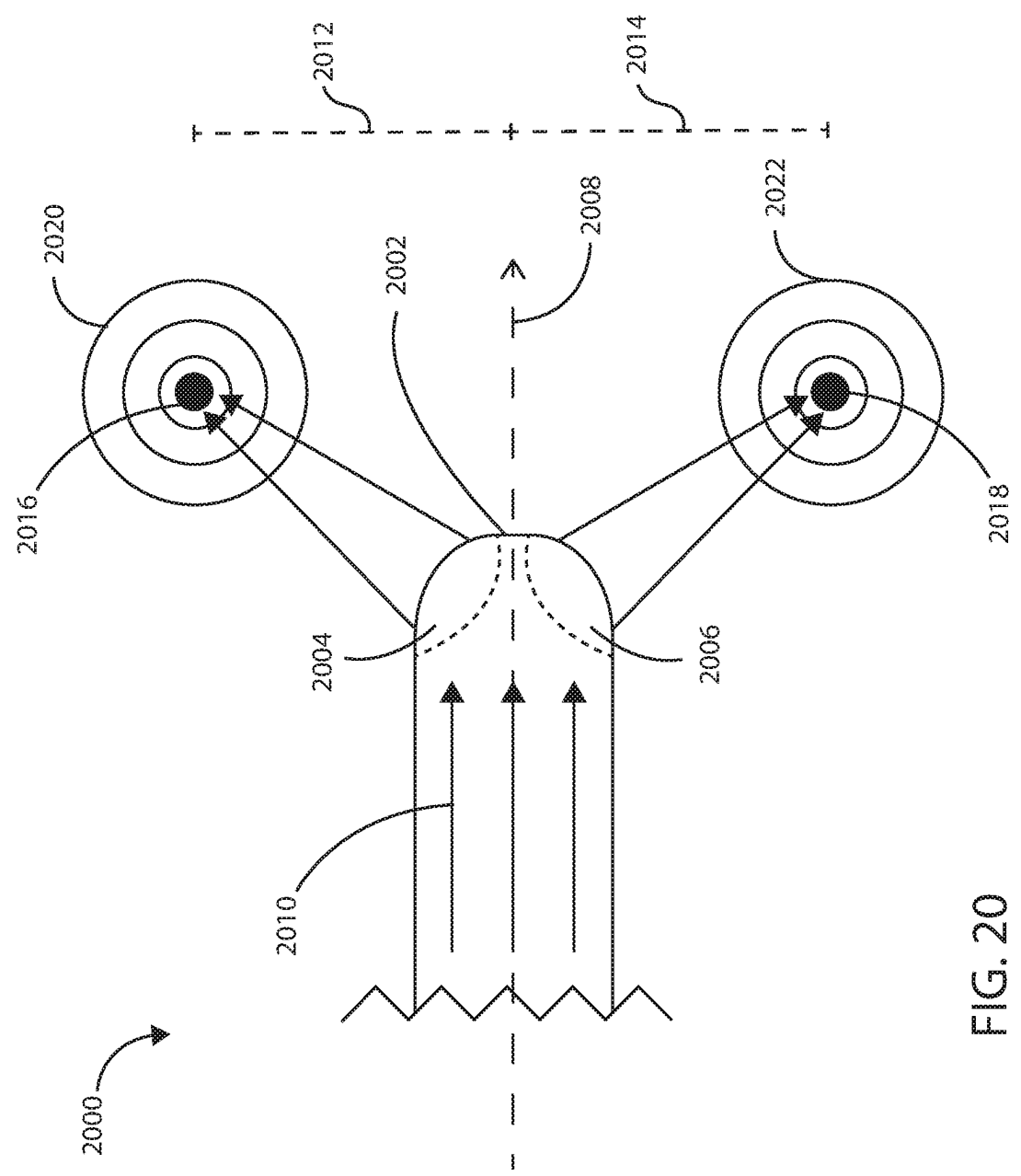


FIG. 20

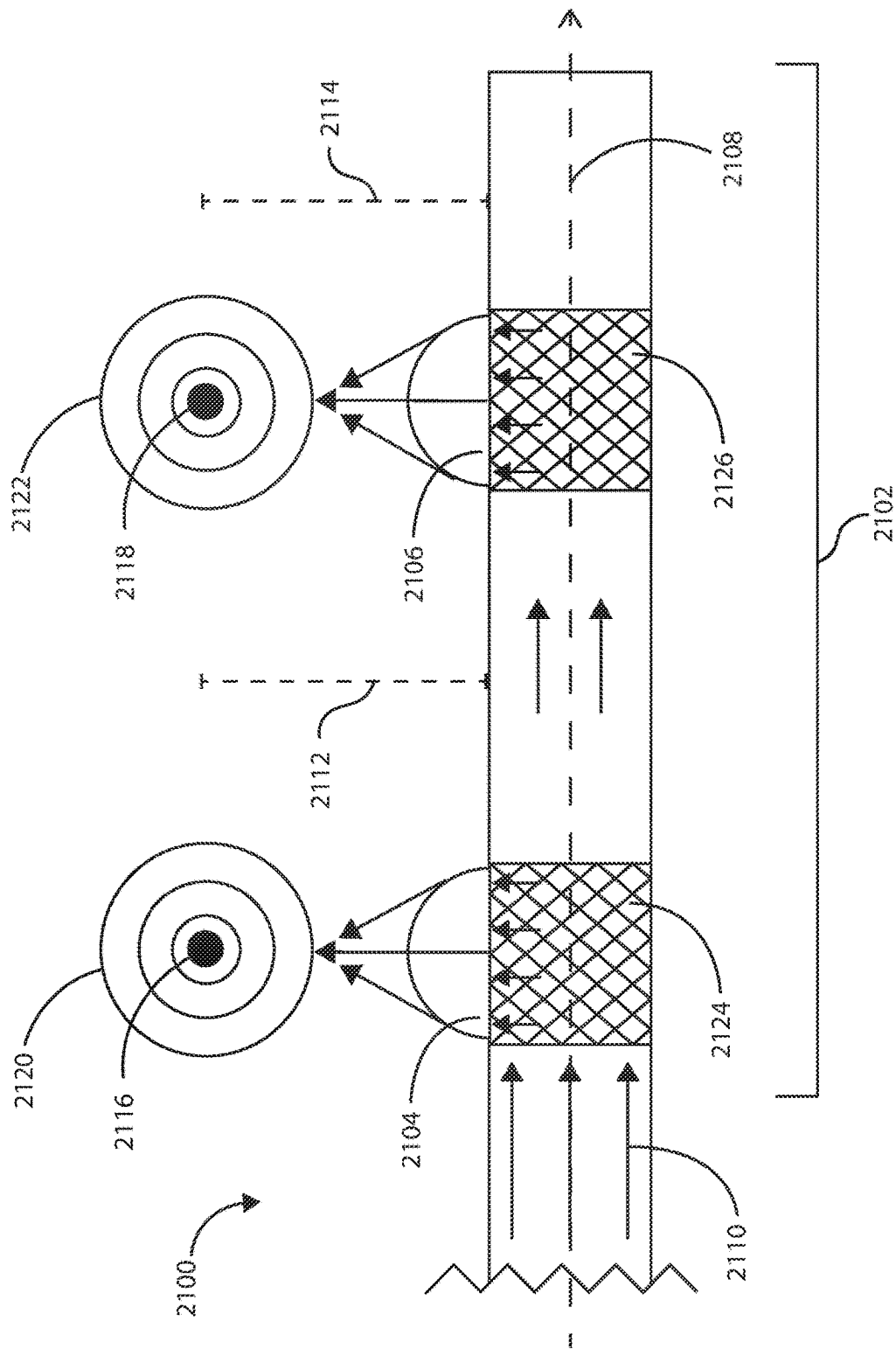


FIG. 21

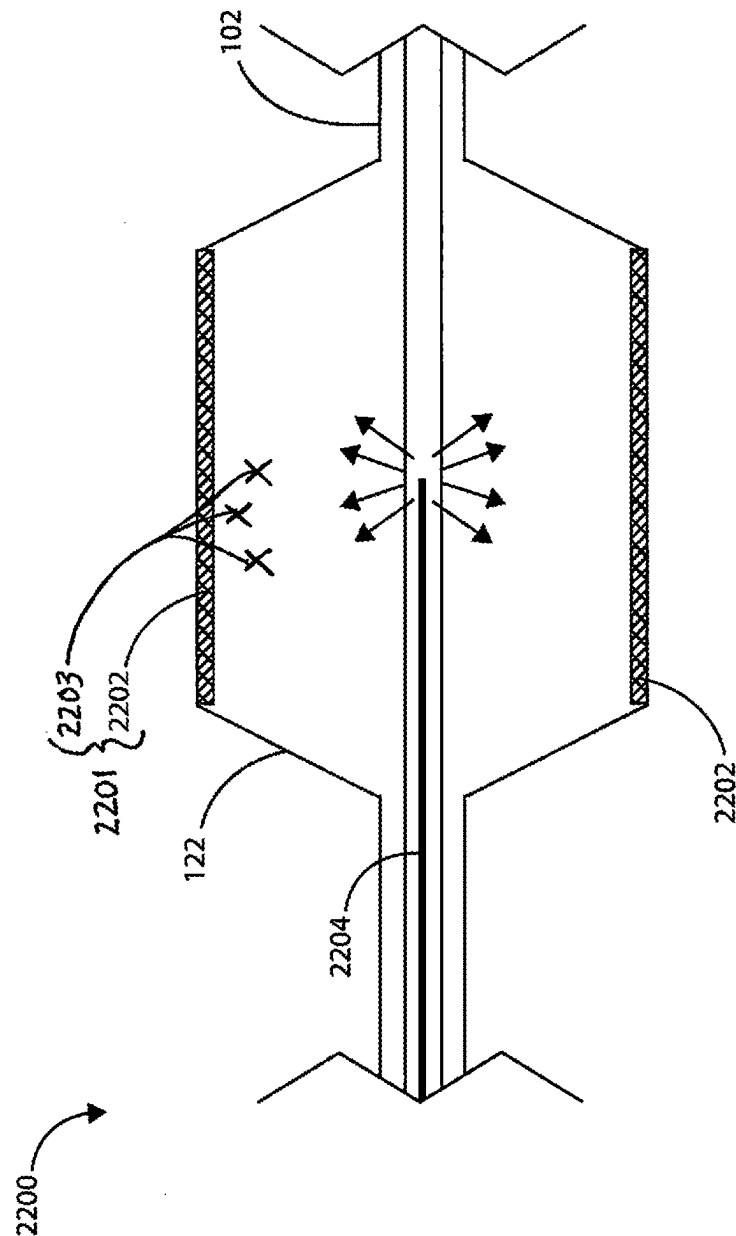


FIG. 22

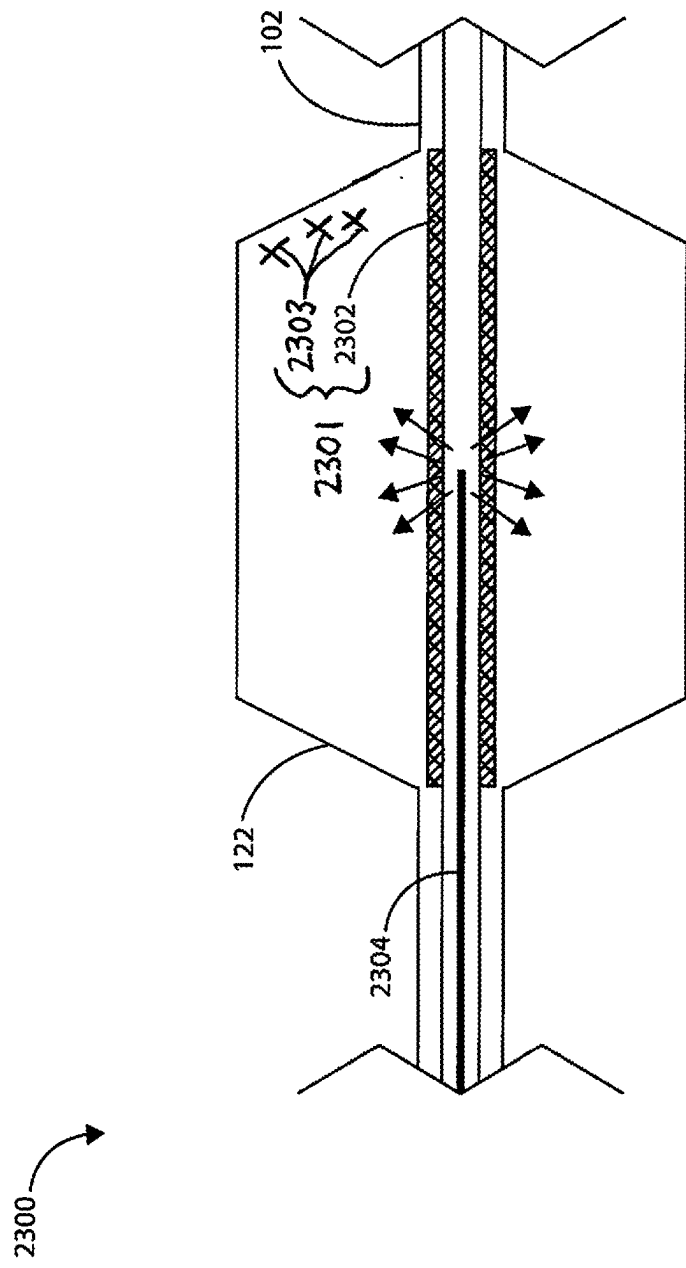


FIG. 23

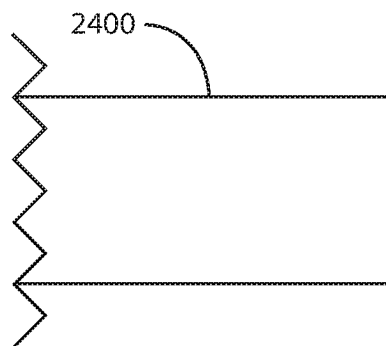


FIG. 24

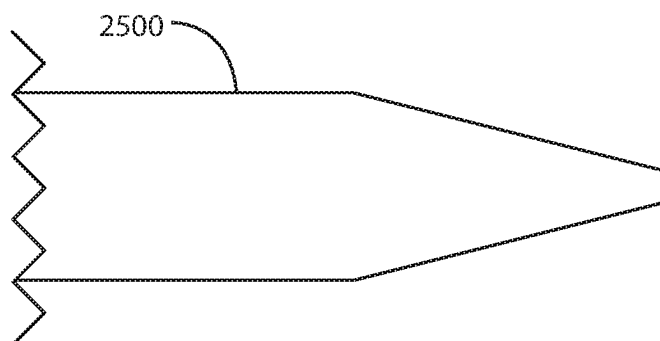


FIG. 25

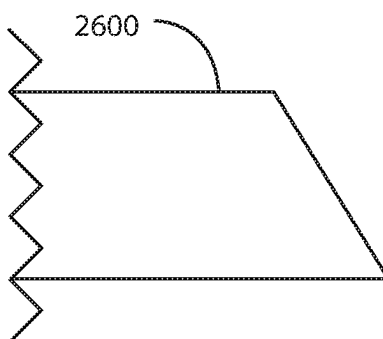


FIG. 26

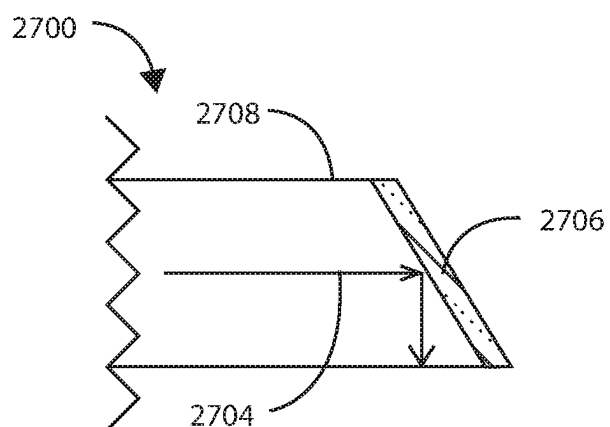


FIG. 27

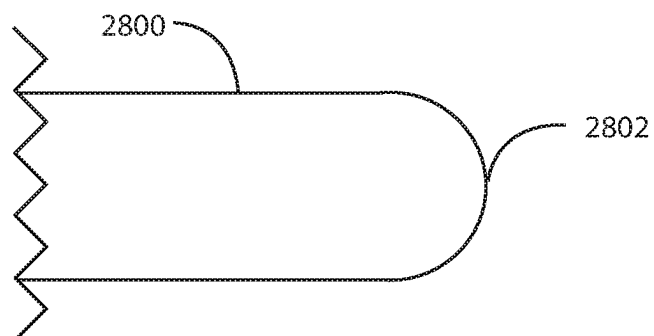


FIG. 28

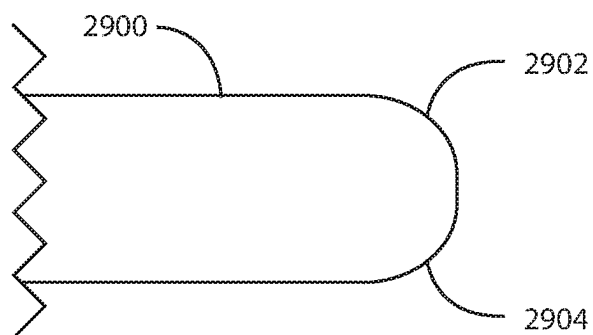
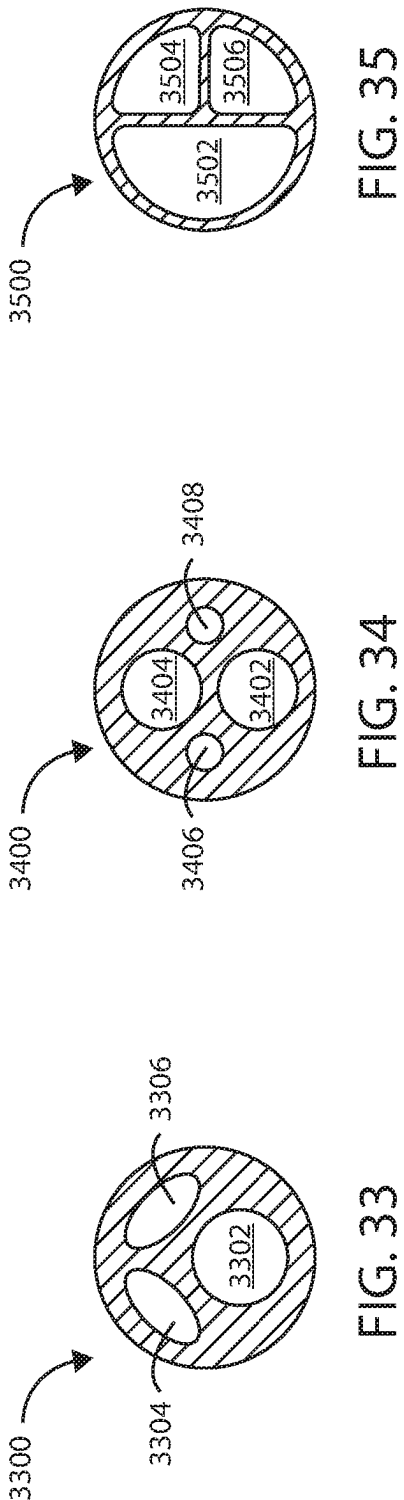
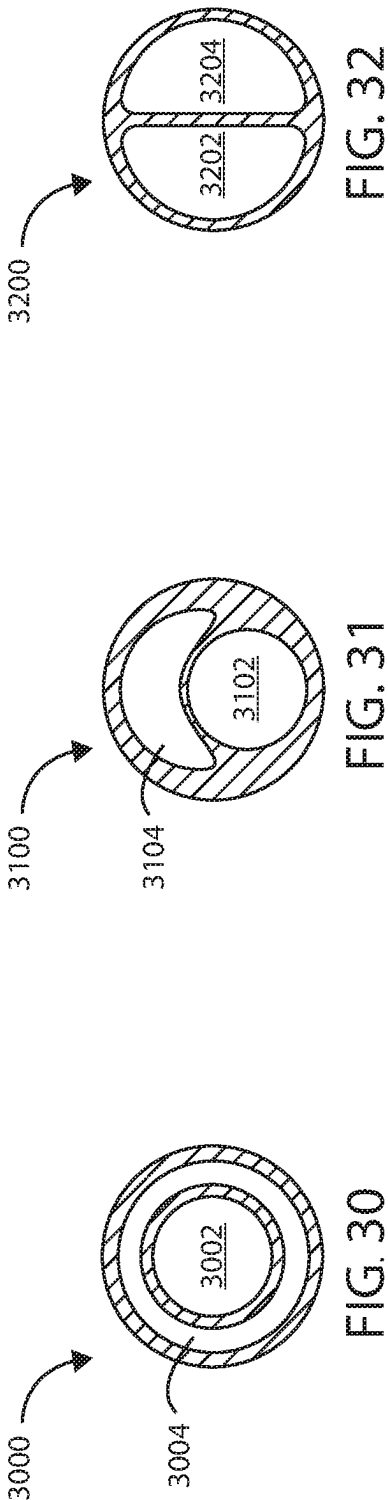


FIG. 29



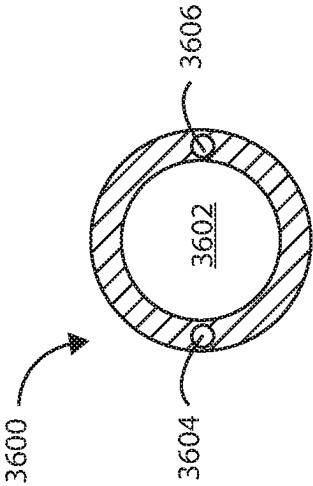


FIG. 36

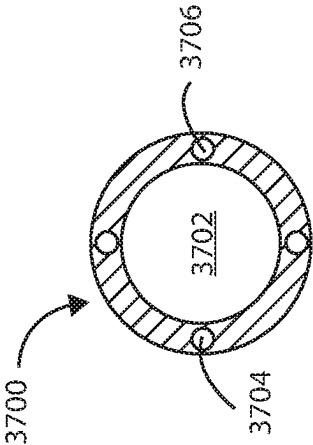


FIG. 37

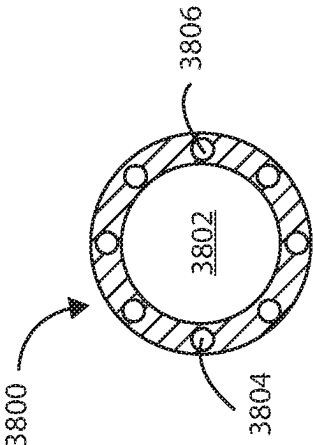


FIG. 38

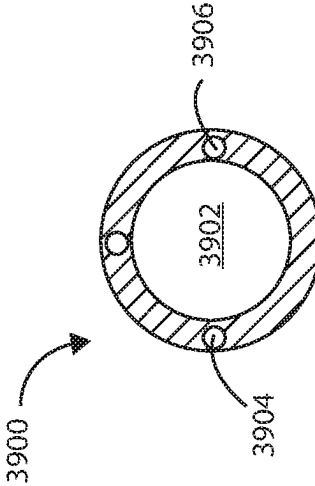


FIG. 39

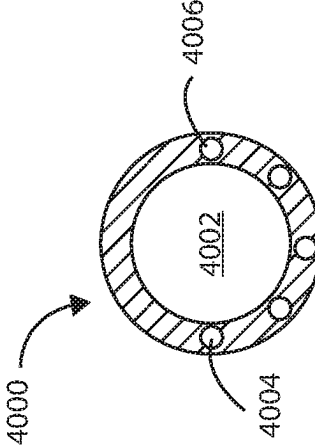


FIG. 40

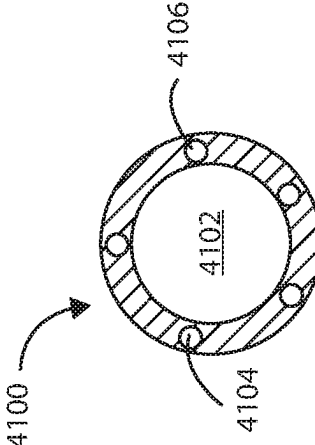


FIG. 41

FORTIFIED BALLOON INFLATION FLUID FOR PLASMA SYSTEM TO DISRUPT VASCULAR LESIONS

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/589,076, filed on Feb. 27, 2024, which is a continuation of U.S. application Ser. No. 16/904,297, filed on Jun. 17, 2020 and issued as U.S. Pat. No. 11,911,574, on Feb. 27, 2024, which claims priority on U.S. Provisional Application Ser. No. 62/866,981, filed on Jun. 26, 2019, and entitled “SIDE LIGHT DIRECTION PLASMA SYSTEM TO DISRUPT VASCULAR LESIONS”, on U.S. Provisional Application No. 62/867,009, filed on Jun. 26, 2019, entitled, “LIGHT GUIDE PROTECTION STRUCTURES FOR PLASMA SYSTEM TO DISRUPT VASCULAR LESIONS”, on U.S. Provisional Application No. 62/867,026, filed on Jun. 26, 2019, entitled, “FORTIFIED BALLOON INFLATION FLUID FOR PLASMA SYSTEM TO DISRUPT VASCULAR LESIONS”, and on U.S. Provisional Application No. 62/867,034, filed on Jun. 26, 2019, entitled, “FOCUSING ELEMENT FOR PLASMA SYSTEM TO DISRUPT VASCULAR LESIONS”, the contents of which for each application are herein incorporated by reference in their entirety to the extent permitted.

BACKGROUND

[0002] Vascular lesions within vessels in the body can be associated with an increased risk for major adverse events, such as myocardial infarction, embolism, deep vein thrombosis, stroke, and the like. Severe vascular lesions can be difficult to treat and achieve patency for a physician in a clinical setting.

[0003] Vascular lesions may be treated using interventions such as drug therapy, balloon angioplasty, atherectomy, stent placement, vascular graft bypass, to name a few. Such interventions may not always be ideal or may require subsequent treatment to address the lesion.

SUMMARY

[0004] In a first aspect, a catheter system for imparting pressure to induce fractures in a treatment site within or adjacent a blood vessel wall is included. The catheter systems can include a catheter configured to advance to the treatment site located within or adjacent a blood vessel. The catheter can include an elongate shaft, and a balloon coupled to the elongate shaft, where the balloon can include a balloon wall. The balloon can be configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a treatment site. The catheter can include a fortified balloon inflation fluid that includes a base inflation fluid and a fortification component. The fortification component can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid, where the fortification component can include carbon or iron. The catheter can include a first light guide disposed along the elongate shaft and within the balloon, where the first light guide can be configured to be placed in optical communication with a light source and the fortified balloon inflation fluid. The light source can be configured to provide sub-millisecond pulses of a light from

the light source to the first light guide so that plasma formation and rapid bubble formation occur in the fortified balloon inflation fluid, thereby imparting pressure waves upon the treatment site.

[0005] In a second aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component can include an iron dextran.

[0006] In a third aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the iron dextran can be present in the fortified balloon inflation fluid at a concentration of from at least 0.0001 millimole per liter (mmol/L) to 1.0 mmol/L.

[0007] In a fourth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component includes nanoparticles.

[0008] In a fifth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component includes iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, gold-coated carbon nanotubes, or copper-coated carbon nanotubes.

[0009] In a sixth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the base inflation fluid includes saline and contrast medium in a ratio of saline to contrast medium of 25:75 volume percent to 75:25 volume percent.

[0010] In a seventh aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component is configured to modify one or more of viscosity, density, or surface tension of the fortified balloon inflation fluid compared to the base inflation fluid.

[0011] In an eighth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the elongate shaft can define an inflation lumen, where the inflation lumen is in fluid communication with the balloon at a distal portion of the elongate shaft and in fluid communication with a fluid source at a proximal end of the elongate shaft.

[0012] In a ninth aspect, a catheter system for imparting pressure to induce fractures in a treatment site within or adjacent a blood vessel wall, is included having a catheter configured to advance to the treatment site located within or adjacent a blood vessel, where the catheter can include an elongate shaft, and a balloon coupled to the elongate shaft, where the balloon can include a balloon wall. The catheter can include a fortification component coating disposed on an inside surface of the balloon wall or on the elongate shaft, where the fortification component includes carbon or iron. The balloon can be configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a treatment site. The catheter can include a base inflation fluid, where the base inflation fluid is configured to solvate the fortification component coating on the inside surface of the balloon wall or elongate shaft to form a fortified balloon inflation fluid. The fortification component can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid. The catheter can include a first light guide disposed along the elongate shaft and within the balloon, where the first light guide can be configured to be placed in

optical communication with a light source and the fortified balloon inflation fluid. The light source can be configured to provide sub-millisecond pulses of a light from the light source to the first light guide so that plasma formation and rapid bubble formation occur in the fortified balloon inflation fluid, thereby imparting pressure waves upon the treatment site.

[0013] In a tenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component includes iron dextran.

[0014] In an eleventh aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the first light guide is an optical fiber and the light source is a laser.

[0015] In a twelfth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the catheter can further a second light guide coupled to the elongate shaft, where the second light guide is in optical communication with the light source and the fortified balloon inflation fluid.

[0016] In a thirteenth aspect, a method for generating pressure waves to induce fractures in a treatment site within or adjacent a vessel wall of a blood vessel is included. The method can include advancing a catheter to a treatment site within the blood vessel, where the catheter can include an elongate shaft, a balloon coupled to the elongate shaft, and at least a first light guide disposed along the elongate shaft within the balloon. The method can include expanding the balloon from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a treatment site. Expanding the balloon can include expanding the balloon with a fortified balloon inflation fluid, where the fortified balloon inflation fluid can include a base inflation fluid and a fortification component. The fortification component can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid, where the fortification component includes carbon or iron. The method can include, after expanding the balloon, activating a light source in optical communication with the light guide and the fortified balloon inflation fluid to provide sub-millisecond pulses of a light from the light source to the fortified balloon inflation fluid so that plasma formation and rapid bubble formation occur in the fortified balloon inflation fluid, thereby imparting pressure waves upon the treatment site.

[0017] In a fourteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, then method further can include, after activating the light source, further expanding the balloon from the first expanded configuration to a second further expanded configuration.

[0018] In a fifteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component includes iron dextran or nanoparticles.

[0019] In a sixteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the fortification component includes nanoparticles, and where the nanoparticles include iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, gold-coated carbon nanotubes, or copper-coated carbon nanotubes.

[0020] In a seventeenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the first light guide includes an optical fiber and the light source includes a laser.

[0021] In an eighteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the sub-millisecond pulses of light from the light source are delivered at a frequency of from at least 1 Hz to 30 Hz.

[0022] In a nineteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the sub-millisecond pulses of light from the light source are delivered at a frequency of from at least 30 Hz to 5000 Hz.

[0023] In a twentieth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where expanding the balloon with the fortified balloon inflation fluid includes providing the fortification component as a coating disposed on an inside surface of a balloon wall or on the elongate shaft in fluid communication with the base inflation fluid, providing the base inflation fluid to the balloon, and the base inflation fluid solvating the fortification component coating to form the fortified balloon inflation fluid.

[0024] This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope herein is defined by the appended claims and their legal equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The novel features of this invention, as well as the invention itself, both as to its structure and its operation, will be best understood from the accompanying drawings, taken in conjunction with the accompanying description, and in which:

[0026] FIG. 1 is a schematic cross-sectional view of a catheter in accordance with various embodiments herein.

[0027] FIG. 2 is a schematic cross-sectional view of an elongate shaft and multiple light guides of a catheter along line 2-2' in FIG. 1 in accordance with various embodiments herein.

[0028] FIGS. 3-5 are schematic cross-sectional views of additional configurations for an elongate shaft and multiple light guides of a catheter along line 2-2' in FIG. 1 in accordance with various embodiments herein.

[0029] FIG. 6 is a schematic cross-sectional view of a light guide in accordance with various embodiments herein.

[0030] FIG. 7 is a schematic cross-sectional view of a light guide in accordance with various embodiments herein.

[0031] FIG. 8 is a schematic cross-sectional view of a light guide in accordance with various embodiments herein.

[0032] FIG. 9 is a schematic cross-sectional view of a light guide in accordance with various embodiments herein.

[0033] FIG. 10 is a schematic flow diagram for a method in accordance with the various embodiments herein.

[0034] FIG. 11 is a schematic side-view of a catheter, with a partial longitudinal cross-sectional view of a balloon, in accordance with various embodiments herein.

[0035] FIGS. 12-13 are schematic side views of an elongate shaft in accordance with various embodiments herein.

[0036] FIGS. 14-15 are schematic cross-sectional views of various distal tip protection structures in accordance with various embodiments herein.

[0037] FIGS. 16-18 are schematic cross-sectional views of various longitudinal protection structures in accordance with various embodiments herein.

[0038] FIG. 19 is a schematic side-view of a catheter, with a partial longitudinal cross-sectional view of a balloon, in accordance with various embodiments herein.

[0039] FIGS. 20-21 are schematic cross-sectional views of various light guides having multiple focusing elements in accordance with various embodiments herein.

[0040] FIGS. 22-23 are cross-sectional views of various catheters including a fortification component coating on a surface within a balloon.

[0041] FIGS. 24-29 are schematic cross-sectional views of various embodiments of a distal portion of a light guide of a catheter in accordance with various embodiments herein.

[0042] FIG. 30-41 are schematic cross-sectional views of additional embodiments of an elongate shaft of a catheter in accordance with various embodiments herein.

[0043] While embodiments are susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings, and will be described in detail. It should be understood, however, that the scope herein is not limited to the particular aspects described. On the contrary, the intention is to cover modifications, equivalents, and alternatives falling within the spirit and scope herein.

DESCRIPTION

[0044] Treatment of vascular lesions can reduce major adverse events or death in affected subjects. A major adverse event is one that can occur anywhere within the body due to the presence of a vascular lesion. Major adverse events can include, but are not limited to major adverse cardiac events, major adverse events in the peripheral or central vasculature, major adverse events in the brain, major adverse events in the musculature, or major adverse events in any of the internal organs.

[0045] The systems and methods disclosed herein describe the use of a catheter systems including any number of light guides for generating pressure waves within a balloon for disrupting intravascular lesions at the treatment site. The catheter systems herein utilize light energy to generate a plasma at or near a light guide disposed in a balloon located at the treatment site, where the treatment site can include a vascular lesion such as a calcified vascular lesion or a fibrous vascular lesion. The plasma formation can initiate a shock-wave and can initiate the rapid formation of one or more bubbles that can rapidly expand to a maximum size and then dissipate through a cavitation event that can launch a shock-wave upon collapse. The rapid expansion of the plasma-induced bubbles can generate one or more pressure waves within a balloon fluid and thereby impart pressure waves upon the treatment site. The pressure waves can transfer mechanical energy through an incompressible balloon fluid to the treatment site to impart a fracture force on an intravascular lesion. Without wishing to be bound by any particular theory, it is believed that the rapid change in balloon fluid momentum upon a balloon wall that is in contact with

an intravascular lesion is transferred to the intravascular lesion to induce fractures to the lesion.

[0046] The catheter systems herein are configured to impart pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall. The catheter systems can include a catheter configured to advance to the vascular lesion located within or adjacent a blood vessel, where the catheters include an elongate shaft. The catheters also include one or more light guides disposed along the elongate shaft and within a balloon. Each light guide can be configured to be placed in optical communication with a light source.

Light Directed Toward a Balloon Wall (FIG. 1)

[0047] The light guides herein can be configured to include one or more diverting features configured to direct light to exit from the light guide toward a side surface portion of the light guide and toward the balloon wall. The diverting features direct light to exit in a direction away from the axis of the light guide, or in an off-axis direction. The light guides can each include one or more light windows disposed along the longitudinal or radial surfaces of each light guide and in optical communication with a diverting feature. The light windows can include a portion of the light guide that allows light to exit the light guide from within the light guide, such as a portion of the light guide lacking a cladding material on or about the light guide. The balloons herein can be coupled to the elongate shaft and can be inflated with a balloon fluid.

[0048] The balloons herein can include a balloon wall and can be configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to the treatment site. The light source can be configured to provide sub-millisecond pulses of a light from the light source to one or more light windows, and thereby initiate plasma formation in a balloon fluid at or near the light windows to cause rapid bubble formation and to impart pressure waves upon the treatment site.

[0049] As used herein, the terms "treatment site", "intravascular lesion" and "vascular lesion" are used interchangeably unless otherwise noted.

[0050] It is appreciated that the catheter systems herein can include many different forms. Referring now to FIG. 1, a schematic cross-sectional view is shown of a catheter in accordance with various embodiments herein. Catheter system 100 is suitable for imparting pressure to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. Catheter system 100 includes a catheter 101. Catheter 101 can be configured to advance to the treatment site within or adjacent a blood vessel. In some embodiments, the treatment site includes a vascular lesion such as a calcified vascular lesion. In other embodiments, the treatment site includes a vascular lesion such as a fibrous vascular lesion.

[0051] The catheter 101 can include an elongate shaft 102 and a balloon 122 coupled to the elongate shaft 102. The elongate shaft 102 can extend from a proximal portion 104 to a distal portion 106, and can also include a lumen 108. The catheter 101 can include a guidewire 126. In some embodiments, the catheter 101 includes a guidewire lumen. The elongate shaft 102 can further include an inflation lumen. Various lumen configurations and their uses will be

discussed in more detail below. In some embodiments, the catheter **101** can have a distal end opening and can accommodate and be tracked over guidewire **126** to the treatment site. In some embodiments, the catheter **101** does not include a guidewire lumen. In embodiments where the elongate shaft **102** does not include a lumen to be accessed by a caregiver, the elongate shaft **102** can be configured to allow the catheter to be steered through a patient's vasculature.

[0052] The elongate shaft **102** of catheter **101** can be coupled to a first light guide **110** and a second light guide (not shown) in optical communication with a light source **116**. The first light guide and second light guide can be disposed along the elongate shaft and within the balloon. It is appreciated that the second light guide of catheter **101** can be offset from first light guide **110** by 180 degrees about the elongate shaft **102** such that it is obstructed by first light guide **110** in FIG. 1. In some embodiments, the first light guide **110** and second light guide can be optical fibers and the light source can be a laser. The light source **116** can be in optical communication with the first light guide **110** and second light guide at a proximal portion **104** of the elongate shaft **102**. In some embodiments, the elongate shaft can be coupled to multiple light guides such as a third light guide and a fourth light guide. The light source **116** can be in optical communication with the third light guide and the fourth light guide at a proximal portion **104** of the elongate shaft **102**. In some embodiments, the elongate shaft can be coupled to more than a fourth light guide.

[0053] The balloon **122** of catheter **101** can include a balloon wall **123** and can expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to the treatment site. Expansion of the balloons herein to various expanded configurations will be discussed in more detail below. The light source **116** of catheter system **100** can be configured to provide sub-millisecond pulses of light from the light source through the at least first light window and second light window, thereby inducing plasma formation in a balloon fluid, causing rapid bubble formation, and imparting pressure waves upon the treatment site. Exemplary plasma-induced bubbles are shown as bubbles **130** in FIG. 1. In some embodiments, the balloon fluid can be a liquid. Suitable balloon fluids for use herein will be discussed in more detail below. In an embodiment, a catheter herein can include a single light guide. The single light guide can include one or more light windows to direct light out the side of the light guide and toward the balloon wall **123**. In one embodiment, the single light guide can include two light windows that can direct light toward the balloon wall **123** in unison.

[0054] The sub-millisecond pulses of light can be delivered to the treatment site at a frequency of from at least 1 hertz (Hz) to 5000 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 30 Hz to 1000 Hz. In other embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 10 Hz to 100 Hz. In yet other embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 1 Hz to 30 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency that can be greater than or equal to 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, 6 Hz, 7 Hz, 8 Hz,

or 9 Hz, 10 Hz, 20 Hz, 30 Hz, 40 Hz, 50 Hz, 60 Hz, 70 Hz, 80 Hz, 90 Hz, 100 Hz, 200 Hz, 300 Hz, 400 Hz, 500 Hz, 600 Hz, 700 Hz, 800 Hz, 900 Hz, 1000 Hz, 1250 Hz, 1500 Hz, 1750 Hz, 2000 Hz, 2250 Hz, 2500 Hz, 2750 Hz, 3000 Hz, 3250 Hz, 3500 Hz, 3750 Hz, 4000 Hz, 4250 Hz, 4500 Hz, 4750 Hz, or 5000 Hz or can be an amount falling within a range between any of the foregoing.

[0055] It is appreciated that the catheters herein can include any number of light guides in optical communication with the light source **116** at the proximal portion **104** and a balloon fluid **124** within balloon **122** at the distal portion **106**. For example, in some embodiments, the catheters herein can include from one light guide to five light guides. In other embodiments, the catheters herein can include from five light guides to fifteen light guides. In yet other embodiments, the catheters herein can include from ten light guides to thirty light guides. The catheters herein can include one, two, three, four, five, six, seven, eight, nine, or ten light guides. The catheters can include 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 light guides. It is appreciated that catheters herein can include any number of light guides that can fall within a range, wherein any of the foregoing numbers can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range. In some embodiments, the catheters herein can include more than 30 light guides. The catheter **101** can further include a manifold **114** at the proximal portion **104** that include one or more proximal end openings that can accommodate one or more light guides, such as first light guide **110**, a guidewire **126**, and/or an inflation conduit **112**. The catheter system **100** can include an inflator **118** configured to provide inflation of the balloon **122**. Suitable balloon inflation pressures for balloon **122** will be described in more detail elsewhere herein.

[0056] Catheter **101** can include a longitudinal length **128**. The catheters herein will have a longitudinal axis along the elongate shaft and short axis about its circumference. The length of the catheters herein can include those having a length of from 50 cm to 175 cm. In some embodiments, the length of the catheters herein can include those having a length of from 100-160 cm. In some embodiments, the length of the catheters herein can include those having a length of 50 cm, 60 cm, 70 cm, 80 cm, 90 cm, 100 cm, 110 cm, 120 cm, 125 cm, 130 cm, 140 cm, 150 cm, 160 cm, 170 cm, or 175 cm. It is appreciated that the catheters herein can have a usable length that can fall within a range, wherein any of the foregoing lengths can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

[0057] Examples of the catheters in accordance with the various embodiments herein include those having multiple light guides disposed about the elongate shaft at different positions around the circumference, as shown in FIGS. 2-5. Referring now to FIG. 2, a schematic cross-sectional view of a catheter **101** of FIG. 1 along line 2-2' in FIG. 1 is shown in accordance with various embodiments herein. Catheter **101** includes an elongate shaft **102**, a first light guide **210** and a second light guide **212** separated by about 180 degrees around the circumference. The first light guide **210** includes a side surface portion **216** that can include any surface portion about the circumference of the first light guide. The second light guide **212** includes a side surface portion **214** that can include any surface portion about the circumference

of the second light guide. In some examples, the side surface portion **214** spans a portion of the circumference of the light guides herein, such that it is less than 360 degrees around the circumference of the light guides, e.g. less than a fully cylindrical side surface portion **214**. In other examples, the side surface portion **214** can span the entire circumference (a full 360 degrees) of the light guides herein such that it is cylindrical. It is appreciated that any light guide described herein can include a side surface portion about a portion or a full circumference of the light guide.

[0058] Referring now to FIGS. 3-5, schematic cross-sectional views of additional configurations for catheters having multiple light guides are shown in accordance with various embodiments herein. The configuration of catheter **300** in FIG. 3 includes an elongate shaft **102**, a first light guide **310**, a second light guide **312**, and a third light guide **314** separated by about 120 degrees around the circumference. The configuration of catheter **400** in FIG. 4 includes an elongate shaft **102**, a first light guide **410**, a second light guide **412**, a third light guide **414**, and a fourth light guide **416** separated by about 90 degrees around the circumference. The configuration of catheter **500** shown in FIG. 5 includes an elongate shaft **102**, a first light guide **510**, a second light guide **512**, a third light guide **514**, a fourth light guide **516**, a fifth light guide **518**, and a sixth light guide **520** separated by about 60 degrees around the circumference. It is appreciated that more than six light guides can be used in the embodiments herein. It will be further appreciated that the light guides can be disposed uniformly or nonuniformly about the elongate shaft.

[0059] The light guides described herein can further include one or more diverting features (not shown in FIG. 1) within the light guide to direct light toward a side surface portion of the distal portion of the light guide and toward the balloon wall **123**. A diverting feature can include any feature of the system herein that diverts light from the light guide away from its axial path toward a side surface portion of the light guide. It is appreciated that a light guide, such as first light guide **110** of FIG. 1, can include a first diverting feature configured to direct light to exit from the first light guide **110** toward a side surface portion of the distal portion **106** of the first light guide **110** and toward the balloon wall **123**. The first light guide **110** can further define at least a first light window (not shown in FIG. 1) disposed along the first light guide and in optical communication with the first fiber diffuser. A catheter having a second light guide can include a second diverting feature within the second light guide that is configured to direct light to exit from the second light guide toward a side surface portion of the distal portion of the second light guide and toward the balloon wall. The second light guide can further define at least a second light window disposed along the second light guide and in optical communication with the second fiber diffuser.

[0060] In some embodiments herein, the light guides can include multiple diverting features. By way of example, each light guide herein can include a first diverting feature, a second diverting feature, a third diverting feature or a fourth diverting feature. In other embodiments, each light guide can include more than four diverting features. The diverting features can be configured to direct light to exit a light guide at a side surface portion thereof toward the balloon wall. In some examples, the diverting feature directs light toward the balloon surface closest to the diverting feature, so that the light does not cross the longitudinal axis

of the catheter on its path to the balloon surface. It is appreciated that the diverting features can be in optical communication with corresponding light window.

[0061] The diverting features herein can be configured to direct light in the light guide toward a side surface portion of the distal portion, where the side surface portion is in optical communication with a light window. It is appreciated that the light guides herein can each include multiple diverting features and multiple light windows. Examples of the diverting features suitable for use herein include a reflecting element, a refracting element, and a fiber diffuser, and will be discussed in more detail below. In some embodiments, the diverting feature can be a reflecting element. In some embodiments, the diverting feature can be a refracting element. In some embodiments, the diverting feature can be a fiber diffuser. Diverting features will be discussed in more detail below and in reference to FIGS. 6-9.

Catheter Embodiments (FIGS. 6-9)

[0062] In some embodiments, the diverting features within the catheters herein can be included within the light guide at one or more regions of the distal portion. Referring now to FIG. 6 a schematic cross-sectional view of light guide **600** is shown in accordance with various embodiments herein. Light guide **600** includes a plurality of diverting features, such as fiber diffusers including first, second, and third fiber diffusers **602**, **604**, and **606**, respectively, positioned along the elongate shaft of the distal portion of the light guide **600**. Each fiber diffuser directs light **601** from the light guide **600** to exit the light guide **600** at a side surface portion **622** thereof. Any side surface portion of the light guide **600** can be in optical communication with one or more light windows, such that the fiber diffusers and the light windows are in optical communication with one another.

[0063] By way of example, light guide **600** includes a plurality of light windows including first, second, and third light windows **608**, **610**, and **612**, respectively, positioned along the elongate shaft of the light guide **600**. The first, second, and third light windows **608**, **610**, and **612**, respectively, can be in optical communication with the first, second, and third fiber diffusers **602**, **604**, and **606**, respectively, at a side surface portion **622** of light guide **600**. Light **601** within each of the first, second, and third fiber diffusers **602**, **604**, and **606** is directed to exit the light guide **600** at a side surface portion **622** and out the light guides via the first, second, and third light windows **608**, **610**, and **612**, respectively. The light windows **608**, **610**, and **612** of light guide **600** can be axially spaced apart with at least one intervening non-emitting portion **620** of the light guide **600** disposed between the plurality of light windows. The side surface portion **622** of the light guide **600** can be a cylindrical side surface portion.

[0064] The light can exit the light windows to provide sub-millisecond pulses of light from the light source through the at least the first, second, and third light windows **608**, **610**, and **612**, thereby inducing plasma formation in a balloon fluid, causing rapid bubble formation, and imparting pressure waves upon the treatment site. Plasma and bubble formation is depicted in FIG. 6 as bubbles **614** in proximity to the light windows **608**, **610**, and **612**. It is appreciated that bubbles **614** can form around the entire circumferences of the light guide within a volume of the balloon fluid at or adjacent to the light windows **608**, **610**, and **612**.

[0065] The fiber diffusers and light windows shown in FIG. 6 include those having a cylindrical shape. By way of example, the fiber diffusers 602, 604, and 606 are configured to span the entire circumference (360 degrees) of light guide 600, and as such, the fiber diffusers 602, 604, and 606 are cylindrical fiber diffusers. The light windows 608, 610, and 612 are configured to span the entire circumference of light guide 600, and as such, light windows 608, 610, and 612 are cylindrical plasma generators. It is appreciated that the cylindrical fiber diffusers 602, 604, and 606 and cylindrical light windows 608, 610, and 612 can generate a plasma within a volume of the balloon fluid at or adjacent to each of the light windows 608, 610, and 612, and thus induce bubble formation and collapse, about the circumference of the light guide 600.

[0066] It is appreciated that multiple light guides, each having one or more diverting features, such as fiber diffusers, and one or more light windows can be used with the catheters herein. In some embodiments, the catheters can include a first light guide, a second light guide, a third light guide, and a fourth light guide. In other embodiments, the catheters can include more than four light guides. In an embodiment having four light guides, the distal portion of a first light guide can include a plurality of light windows including a first light window, and a plurality of fiber diffusers including a first fiber diffuser. The distal portion of a second light guide can include a plurality of light windows including a second light window, and a plurality of fiber diffusers including the second fiber diffuser. The distal portion of a third light guide can include a plurality of light windows including a third light window, and a plurality of fiber diffusers including a third fiber diffuser. The distal portion of a fourth light guide can include a plurality of light windows including a fourth light window, and a plurality of fiber diffusers including the fourth fiber diffuser. Each fiber diffuser can direct light from each light guide to exit the light guide at a side surface portion of the light guide toward the balloon wall.

[0067] The plurality of light windows can be spaced apart along the longitudinal axis of the light guides or radially along the short axis about the circumference of the light guides. In some embodiments, the plurality of light windows can be radially spaced apart with at least one intervening non-emitting portion of the light guide disposed between each of the plurality of light windows. In some embodiments, the plurality of light windows can be longitudinally spaced apart with at least one intervening non-emitting portion of the light guide disposed between each of the plurality of light windows. In yet other embodiments, the light window can span the length of the vascular lesion to be treated.

[0068] In some embodiments, the catheters herein can include diverting features, such as fiber diffusers, in combination with one or more focusing elements included within the light guide at one or more regions of the distal portion. Referring now to FIG. 7 a schematic cross-sectional view of light guide 700 is shown in accordance with various embodiments herein. Light guide 700 includes a plurality of fiber diffusers including a first fiber diffuser 702 and a second fiber diffuser 704 positioned along the longitudinal axis of the distal portion of the light guide 700. Each fiber diffuser directs light 701 from the light guide 700 to exit the light

guide 700 at a side surface portion 722 thereof. The side surface portion 722 of the light guide 700 can be a cylindrical side surface portion.

[0069] The first fiber diffuser 702 can be in optical communication with a first light window 706, and the second fiber diffuser 704 can be in optical communication with a second light window 708. The light guide 700 can further include a refracting element 710 configured to focus the light 701 away from the distal tip of the light guide 700 such that the induced plasma formation occurs at a distance 712 away from the distal tip of the light guide 700 and within the balloon fluid, causing rapid bubble formation and imparting pressure waves at the treatment site. The light 701 within light guide 700 can exit the first light window 706 and the second light window 708 to deliver sub-millisecond pulses of light from the light source thereby inducing plasma formation in a volume of balloon fluid at or near the first light window 706 and second light window 708, causing rapid bubble formation, and imparting pressure waves upon the treatment site. Plasma and bubble formation is depicted in FIG. 7 as bubbles 714 in proximity to the light windows 706 and 708.

[0070] The fiber diffusers and light windows shown in FIG. 7 include those having a cylindrical shape. By way of example, the fiber diffusers 702 and 704 are configured to span the entire circumference (360 degrees) of light guide 700, and as such, the fiber diffusers 702 and 704 are therefore cylindrical fiber diffusers. The light windows 706 and 708 can be configured to span the entire circumference (360 degrees) of light guide 700, and as such, light windows 706 and 708 would be cylindrical plasma generators. It is appreciated that the cylindrical fiber diffusers 702 and 704 and cylindrical light windows 706 and 708 can generate plasma, and thus bubble formation, about the circumference, or a portion of the circumference, of the light guide 700.

[0071] The light guide 700 shown in FIG. 7 also includes a diverting feature, such as a refracting element 710 having a convex surface configured to refract the light 701 a distance 712 away from the distal portion of the light guide 700 and to a first location 718. In some embodiments, the first location 718 is spaced away from the distal tip and is centered on a longitudinal axis of the first light guide. In other embodiments, the first location 718 is spaced away from the longitudinal axis of the first light guide, or off of the longitudinal axis. The diverting feature suitable for focusing light away from the tip of the light guides herein can include, but are not to be limited to, those having a convex surface, a gradient-index (GRIN) lens, and a mirror focus lens. Focused plasma formation is depicted in FIG. 7 as focal bubble 716 at a distance 712 from the distal portion of the light guide 700.

[0072] The light guides herein can include one or more diverting features disposed on one side portion of the distal portion to provide multiple selected regions within the light guide for the generation of pressure waves. A diverting feature can be included as a part of the light guide that diverts light away from its axial path through the light guide and to a side surface portion and toward a vessel wall. Referring now to FIG. 8 a schematic cross-sectional view of light guide 800 is shown in accordance with various embodiments herein. Light guide 800 includes a plurality of diverting features including a first, second, and third diverting features 802, 804, and 806 positioned along a portion of the elongate shaft of the distal portion of the light guide 800.

Each diverting feature directs light **801** from the light guide **800** to exit the light guide **800** at a side surface portion **822** thereof. The side surface portion **822** of the light guide **800** can be in optical communication with one or more diverting features and one or more light windows, such that the diverting features and one or more light windows are in optical communication with one another.

[0073] By way of example, light guide **800** includes a plurality of light windows including a first, second, and third light windows **808**, **810**, and **812** positioned along the elongate shaft of the light guide **800**. The first, second, and third light windows **808**, **810**, and **812** can be in optical communication with the first, second, and third diverting features **802**, **804**, and **806**, respectively, at a side surface portion **822** of light guide **800**. Light within each of the first, second, and third diverting features **802**, **804**, and **806** is directed to exit the light guide **800** at a side surface portion **822** and pass through first, second, and third light windows **808**, **810**, and **812**, respectively. The light can exit the light windows to provide sub-millisecond pulses of light from the light source through the at least first light window and second light window, thereby inducing plasma formation in a balloon fluid, causing rapid bubble formation, and imparting pressure waves upon the treatment site. Plasma and bubble formation is depicted in FIG. 8 as bubbles **814** in proximity to the light windows **808**, **810**, and **812**. The light windows **808**, **810**, and **812** of light guide **800** can be longitudinally spaced apart with at least one intervening non-emitting portion **820** of the light guide **800** disposed between the plurality of light windows.

[0074] In various examples, the light windows, diverting features, or both can vary in size and shape along the length of the catheter. In various examples, the light windows, diverting features, or both can be dome-shaped, square, triangular, circular, rectangular, and the like, and can increase in size moving toward the distal portion. In the example of FIG. 8, the most proximal first diverting feature **802** is smaller than the second diverting feature **804**, and the third diverting feature **806** is larger than the second diverting feature **804**. In the example of FIG. 8, the most proximal first light window **808** is smaller in surface area than the second light window **810**, and the third light window **812** is larger in surface area than the second light window **810**.

[0075] While the light windows and diverting features of light guide **800** are shown disposed on one side portion of light guide **800**, it is appreciated that the light windows and diverting features can be disposed in many different positions along the elongate shaft. In various examples, light windows and diverting features can be disposed opposite one another along the elongate shaft of the light guide. Referring now to FIG. 9, a schematic cross-sectional view of light guide **900** is shown in accordance with various embodiments herein. Light guide **900** includes diverting features **902**, **904**, **906**, and **908**. Each diverting feature directs light **901** from the light guide **900** to exit the light guide **900** at a side surface portion **922** thereof. The side surface portion **922** of the light guide **900** can be a cylindrical side surface portion. The side surface portion **922** of the light guide **900** can be in optical communication with one or more diverting features and one or more light windows, such that the diverting features and light windows are in optical communication with one another.

[0076] By way of example, light guide **900** includes a plurality of light windows including a first, second, third,

and fourth light windows **910**, **912**, **916**, and **918**, respectively, positioned along the elongate shaft of the light guide **900**. The first, second, third, and fourth light windows **910**, **912**, **916**, and **918**, respectively, can be in optical communication with the first, second, third, and fourth diverting features **902**, **904**, **906**, and **908**, respectively, at a plurality of side surface portion **922** of light guide **900**. Light within each of the first, second, third, and fourth diverting features **902**, **904**, **906**, and **908** is directed to exit the light guide **900** at a side surface portion **922** and exits through the first, second, third, and fourth light windows **910**, **912**, **916**, and **918**, respectively. Light energy can exit light windows **910**, **912**, **916**, and **918** and induce plasma formation in a volume of balloon fluid at or near the light windows **910**, **912**, **916**, and **918**, causing rapid bubble formation, and imparting pressure waves upon the treatment site. Plasma and bubble formation is depicted in FIG. 9 as bubbles **914** in proximity to the light windows **910**, **912**, **916**, and **918**. The light windows **910**, **912**, **916**, and **918** of light guide **900** can be longitudinally spaced apart with at least one intervening non-emitting portion **920** of the light guide **900** disposed between the plurality of light windows.

[0077] The catheters described herein can be used in one or more methods for generating pressure waves to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. Referring now to FIG. 10, a schematic flow diagram for a method **1000** is shown in accordance with the various embodiments herein. Method **1000** includes advancing a catheter **1010** to a treatment site **1014** within the blood vessel **1012**, the catheter **1010** including an elongate shaft **102**, and a balloon **122** coupled to the elongate shaft **102** at **1002**. In some embodiments, the treatment site **1014** can include a vascular lesion location within a patient's vasculature. In some embodiments, the vascular lesion can include a calcified vascular lesion or a fibrous vascular lesion.

[0078] The method **1000** includes expanding the balloon **122** from a collapsed configuration suitable for advancing the catheter **1010** through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to the treatment site **1014** at **1004**. The method **1000** includes, after expanding the balloon **122**, activating a light source in optical communication with each of the first light guide and the second light guide to provide sub-millisecond pulses of light from the light source to the at least first diverting feature and second diverting feature, thereby initiating plasma formation in a balloon fluid, causing rapid bubble formation, and imparting pressure waves **1016** upon the treatment site **1014** at **1004**.

[0079] In some embodiments, the method **1000** includes a first light guide having a first diverting feature configured to direct light to exit from the first light guide toward a side surface portion of the distal portion of the first light guide and toward the balloon wall, where the first light guide defines a first light window in optical communication with the first diverting feature.

[0080] In some embodiments, the method **1000** includes a second light guide having a second diverting feature configured to direct light to exit from the second light guide toward a side surface portion of the distal portion of the second light guide and toward the balloon wall, where the second light guide defines a second light window in optical communication with the second diverting feature.

[0081] In some embodiments, the method **1000** includes a third light guide having a third diverting feature configured to direct light to exit from the third light guide toward a side surface portion of the distal portion of the third light guide and toward the balloon wall, where the third light guide defines a third light window in optical communication with the third diverting feature.

[0082] In some embodiments, the method **1000** includes a fourth light guide having a fourth diverting feature configured to direct light to exit from the fourth light guide toward a side surface portion of the distal portion of the fourth light guide and toward the balloon wall, where the fourth light guide defines a fourth light window in optical communication with the fourth diverting feature. In some embodiments, the method **1000** includes more than four light guides.

[0083] The method **1000** can also include further expanding the balloon **122** from the first expanded configuration to a second further expanded configuration at **1006**. The method can include completely removing the catheter **1010** from the patient's vasculature at **1008**.

[0084] The light guides and components associated therewith that are suitable to be used in the methods herein can be activated in various ways to provide a treatment to the treatment site. In some embodiments, each light guide can be activated simultaneously. In some embodiments, each light guide can be activated sequentially. By way of example, if two light guides are present, they can each be activated at the same time, they can be activated one after the other sequentially, or they can be activated in alternating pairs, semi-randomly, randomly or in another suitable fashion. The light guides can be activated once or multiple times during the course of a treatment. In an embodiment with four light guides, each of the four light guides can be activated at the same time, sequentially, in pairs, or in alternating pairs. By way of example, if four light guides are present, each with one light window, the first and third light guide and their respective light windows can form a pair that can be activated followed by activation of the second and fourth light guide and their respective light windows that can form a pair, either once each or in an ongoing alternating fashion. It is appreciated that many configurations exist for activating multiple light guides and their respective light windows in accordance with the embodiments herein.

[0085] Vascular lesions can be present within or about a vessel wall of a blood vessel in various configurations, including surrounding the entire lumen of the vessel or surrounding a portion of the lumen of a vessel. Vascular lesions can also be present in various shapes and sizes. To provide targeted therapy to a vascular lesion, the light guides herein can be configured to be activated depending on the vascular lesion location, shape, and size. By way of example, if a vascular lesion is located partially about the circumference of a blood vessel, the light guides can be activated partially about the circumference of the catheter to match the location, size, and shape of the vascular lesion. In some embodiments, where the vascular lesion spans the entire circumference of the blood vessel, the light guides herein can be activated about the entire circumference of the blood vessel. In various embodiments, the light guides can additionally be activated to match the length and width of the vascular lesion.

[0086] The duration of the methods herein can vary according to the specific treatment site and size of a vascular lesion. In some embodiments, the total treatment time can be

from one second to one-hundred twenty seconds. In some embodiments, the total treatment time can be from five seconds to twenty seconds. In other embodiments, the total treatment time can be from five seconds to ten seconds. Still alternatively, the total treatment time can fall within or outside of the foregoing ranges, which are not intended to be limiting.

[0087] The sub-millisecond pulses of light can be delivered to the treatment site at a frequency of from at least 1 hertz (Hz) to 5000 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 30 Hz to 1000 Hz. In other embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 10 Hz to 100 Hz. In yet other embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency from at least 1 Hz to 30 Hz. In some embodiments, the sub-millisecond pulses of light can be delivered to the treatment site at a frequency that can be greater than or equal to 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, 6 Hz, 7 Hz, 8 Hz, or 9 Hz, 10 Hz, 20 Hz, 30 Hz, 40 Hz, 50 Hz, 60 Hz, 70 Hz, 80 Hz, 90 Hz, 100 Hz, 200 Hz, 300 Hz, 400 Hz, 500 Hz, 600 Hz, 700 Hz, 800 Hz, 900 Hz, 1000 Hz, 1250 Hz, 1500 Hz, 1750 Hz, 2000 Hz, 2250 Hz, 2500 Hz, 2750 Hz, 3000 Hz, 3250 Hz, 3500 Hz, 3750 Hz, 4000 Hz, 4250 Hz, 4500 Hz, 4750 Hz, or 5000 Hz or can be an amount falling within a range between any of the foregoing.

Optical Fiber Protection Configurations (FIGS. 11-18)

[0088] The catheter systems herein can include various configurations that include one or more protection structures configured to provide structural protection to one or more portions of the light guides herein when in the presence of the pressure waves. The catheter systems utilizing protection structures can be suitable for imparting pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall. Such catheter systems can include a catheter configured to advance to the vascular lesion located within or adjacent a blood vessel. In various configurations the catheters can include an elongate shaft and a balloon coupled to the elongate shaft. The balloon can be configured to be filled with balloon fluid and configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to the treatment site. In the embodiments described herein, the elongate shaft can define a first recess. In some embodiments, the first recess can include a longitudinal recess along a longitudinal surface of the elongate shaft. In some embodiments, the first recess can include a distal tip recess along a longitudinal surface of the elongate shaft. Referring now to FIG. 11, a schematic side-view of a catheter **1100** including a protection structure, with a partial longitudinal cross-sectional view of a balloon **122**, is shown in accordance with various embodiments herein.

[0089] The catheter **1100** can include a first light guide **1102** disposed along the elongate shaft **102** and within the balloon **122**. The first light guide **1102** can be configured to be placed in optical communication with a light source and a balloon fluid. It is appreciated that the first light guide **1102** can include an optical fiber and the light source can include a laser, both of which are described in more detail elsewhere herein. The light source in communication with the light

guides herein can be configured to provide pulses of light **1112** to the balloon fluid, thereby initiating plasma formation in the balloon fluid, causing rapid bubble formation of bubble **1114**, and imparting pressure waves upon the vascular lesion. The first light guide **1102** can include a longitudinal axis **1108**.

[0090] A first portion **1104** of the first light guide **1102** can extend into the first recess **1106**. In some embodiments, the first portion **1104** includes a distal tip **1116** of the first light guide **1102**. The catheter **1100** can include a first protection structure **1110** disposed within the first recess **1106** of the elongate shaft **102** and in contact with the first portion **1104** of the light guide **1102**. The first protection structure **1110** can be configured to provide structural protection to the first portion **1104** of the first light guide **1102**. In some embodiments, the first protection structure **1110** can be configured to provide structural protection to the first portion **1104**, when the first portion **1104** is a distal tip **1116** of a light guide. In other embodiments, the first protection structure **1110** can be configured to provide structural protection to the first portion **1104**, when the first portion **1104** is a longitudinal portion of a light guide. The first protection structure **1110** can be configured to provide structural protection to the first portion **1104** of the first light guide **1102** in the presence of the pressure waves. In some examples, the first protection structure **1110** includes a potting material filling the first recess **1106**, where the potting material is optically matched to the first light guide **1102**. It is appreciated that in some embodiments, the potting material filling the first recess **1106** is not optically matched and can serve as a diverting feature.

[0091] As the term is used herein, two materials are “optically matched” if those materials have indices of refraction that are the same when expressed to two decimal places. In other embodiments herein, two materials are not optically matched, or are “optically mismatched”, when the two materials that have indices of refraction that are different when expressed to two decimal places.

[0092] The light guide **1102** of catheter **1100** can further include a first diverting feature **1118** selected from the group including at least one of a reflecting element, a refracting element, or a fiber diffuser, as will be discussed elsewhere herein. In various embodiments, the diverting feature can include a fiber diffuser selected from a group including of a machined portion of the light guide, a laser-machined portion of the light guide, fiber Bragg gratings, a fused splicing forming at least one internal mirror, and a splicing of two or more diffuse regions. In some embodiments, diverting feature **1118** of catheter **1100** can include a fused splicing forming at least one internal mirror, where the first light guide can include a first longitudinal light window **1120** disposed along a longitudinal length of light guide and in optical communication with the first diverting feature **1118**.

[0093] It is appreciated that the first portion of the light guides herein that extend into the recess defined by the elongate shaft can include a part of the light guide that extends into the recess, the entire portion of the light guide that extends into recess, or any portion that overlaps with or is present within the recess. In some examples, multiple portions of the light guides herein can extend into more than one recess along the elongate shaft to form a second portion, a third portion, a fourth portion, and the like, of the light guide present within a second recess, a third recess, a fourth recess, and the like.

[0094] Multiple recesses defined along the longitudinal length of the catheters herein can be suitable for use with one or more light guides disposed along the length of the elongate shaft to provide multiple locations where light can be directed from the light guides. Referring now to FIGS. **12** and **13**, schematic side-views of a portion of the elongate shaft of catheters **1200** and **1300** are shown in accordance to various embodiments herein. In the configuration shown in FIG. **12**, elongate shaft **102** of catheter **1200** includes a first light guide **1202** and a second light guide **1204**. The elongate shaft **102** defines first recesses **1210** and **1212**, and second recesses **1206** and **1208**, along the longitudinal length. First light guide **1202** extends into second recess **1206** and first recess **1210**, such that a first portion **1216** of the first light guide **1202** extends into first recess **1210** and a second portion **1214** of the first light guide **1202** extends into second recess **1206**. Second light guide **1204** extends into first recess **1212** and second recess **1208**, such that a first portion **1220** of the second light guide **1204** extends into first recess **1212** and a second portion **1218** of the second light guide **1204** extends into second recess **1208**.

[0095] The second portion **1214** of the first light guide **1202** and the second portion **1218** of the second light guide **1204** each extend through the entire length of the respective second recesses **1206** and **1208**. The first light guide **1202** and the second light guide **1204** each extend into multiple recesses. By way of example, the first portion **1216** of the first light guide **1202** and the first portion **1220** of the second light guide **1204** each extend partially into the recesses first **1210** and **1212** such that the first portion **1216** of the first light guide **1202** includes a distal tip of the first light guide **1202** and the first portion **1220** of the second light guide **1204** includes a distal tip of the second light guide **1204**. It is appreciated that the first light guide **1202** and the second light guide **1204** are depicted as being disposed 180 degrees about the circumference of the elongate shaft, however, the first light guide **1202** and the second light guide **1204** can be disposed about the elongate shaft in many configurations as discussed elsewhere herein.

[0096] In the configuration in FIG. **13**, elongate shaft **102** of catheter **1300** includes a first light guide **1302**, a second light guide **1304**, and a third light guide **1305**. While not shown in FIG. **13**, it is appreciated that catheter **1300** can further include a fourth light guide disposed 180 degrees about the circumference of the elongate shaft **102** and opposite the third light guide **1305**. In various embodiments, catheter **1300** can include more than four light guides. The elongate shaft **102** of catheter **1300** defines recesses **1306**, **1308**, and **1310**, along the longitudinal length. First light guide **1302** extends into first recess **1306** such that a first portion **1314** of the first light guide **1302** extends into first recess **1306**. Second light guide **1304** extends into second recess **1308** such that a first portion **1316** of the second light guide **1304** extends into second recess **1308**. Third light guide **1305** extends into third recess **1310** such that a first portion **1318** of the third light guide **1305** extends into third recess **1310**. The first portions **1314**, **1316**, and **1318** of the first, second, and third light guide each extend partially into their respective recesses **1306**, **1308**, and **1310**. Each of the first portions **1314**, **1316**, and **1318** of the first, second, and third light guides **1302**, **1304**, and **1305** includes a distal tip portion within each recess.

[0097] The protection structures suitable for use herein can be tailored to various configurations dependent on if the

portion of the light guide to be protected is a distal tip or a longitudinal portion. Referring now to the embodiments in FIGS. 14-18, examples of several side views of an elongate shaft, with longitudinal cross-sectional views of the protection structures, are shown in accordance with various embodiments herein. In the configuration in FIG. 14, catheter 1400 includes a first portion 1404 of a first light guide 1402 that extends into a first recess 1406 defined by the elongate shaft 102 (illustrated in FIG. 1). The catheter 1400 can include a first protection structure 1410 disposed within the first recess 1406 of the elongate shaft 102 and in contact with the first portion 1404 of the light guide 1402. The first protection structure 1410 can be an end cap disposed about the distal portion 1414 of the first portion 1404 of the light guide 1402. The end cap can be flush with a distal tip 1408 of the light guide 1402. The first protection structure 1410 can be adhered to an outer surface of the distal portion 1414 of the first portion 1404 of the light guide 1402 with an adhesive 1412, where the adhesive 1412 and the end cap protection structure 1410 are optically matched to the first light guide 1402. It is appreciated that in some embodiments, the end cap protection structure is not optically matched.

[0098] In the configuration in FIG. 15, catheter 1500 includes a first portion 1504 of a first light guide 1502 that extends into a first recess 1506 defined by the elongate shaft 102 (illustrated in FIG. 1). The catheter 1500 can include a first protection structure 1510 disposed within the first recess 1506 of the elongate shaft 102 and in contact with the first portion 1504 of the light guide 1502. The first protection structure 1510 can be a first component that is abutted against and fused to the distal tip 1508 of the first light guide 1502, where the first component protection structure 1510 is optically matched to the first light guide 1502. In various embodiments, the first component protection structure can be made from glass, quartz, sapphire, diamond, and the like. It is appreciated that in some embodiments, the first component protection structure is not optically matched.

[0099] In the configuration in FIG. 16, catheter 1600 includes a second portion 1604 of a first light guide 1602 that extends into a second recess 1606 defined by the elongate shaft 102 (illustrated in FIG. 1). The second recess 1606 can be disposed within the balloon along a longitudinal surface 1612 of the elongate shaft 102. A second portion 1604 of the first light guide 1602 can extend into or through the second recess 1606, where the second portion 1604 can define a longitudinal light window 1608 disposed along a longitudinal length of the second portion 1604 and in optical communication with a first diverting feature 1614. The catheter 1600 can further include a longitudinal protection structure 1610 in contact with the second portion 1604 of the first light guide 1602 and configured to provide structural protection to the second portion 1604 in the presence of pressure waves. The longitudinal protection structure 1610 can be a sleeve disposed about the longitudinal light window 1608 and can be adhered to the longitudinal surface of the second portion 1604 of the first light guide 1602 with an adhesive (not shown), where the adhesive and the sleeve longitudinal protection structure 1610 are optically matched to the first light guide 1602. In some embodiments, the longitudinal protection structure 1610 and the adhesive are not optically matched.

[0100] In the configuration in FIG. 17, catheter 1700 includes a second portion 1704 of a first light guide 1702

that extends into or through a second recess 1706 defined by the elongate shaft 102. The second recess 1706 can be disposed within a balloon along a longitudinal surface 1712 of the elongate shaft 102. A second portion 1704 of the first light guide 1702 can extend into or through the second recess 1706, where the second portion 1704 can define a longitudinal light window 1708 disposed along a longitudinal length of the second portion 1704 and in optical communication with a first diverting feature 1714. The catheter 1700 can further include a longitudinal protection structure 1710 in contact with the second portion 1704 of the first light guide 1702 and configured to provide structural protection to the second portion 1704 in the presence of pressure waves. The longitudinal protection structure 1710 can be a potting material disposed within the second recess 1706 and around the second portion 1704 of the first light guide 1702, where the potting material longitudinal protection structure 1710 is optically matched to the first light guide 1702. It is appreciated that in some embodiments, the potting material of the longitudinal protection structure 1710 is not optically matched and can serve as a diverting feature.

[0101] In the configuration in FIG. 18, catheter 1800 includes a proximal portion 1804 of a first light guide 1802 that extends into a proximal recess 1806 defined by the elongate shaft 102, and also includes a proximal tip 1816 of a continuation of the first light guide 1808. The proximal recess 1806 can be disposed within a balloon along a longitudinal surface 1812 of the elongate shaft 102. The catheter 1800 can include a longitudinal protection structure 1810 in contact with the distal tip 1814 of a proximal portion 1804 of the first light guide 1802 and the proximal tip 1816 of the continuation of the first light guide 1808. The longitudinal protection structure 1810 can be a first component fused between the distal tip 1814 of the proximal portion 1804 of the first light guide 1802 and the proximal tip 1816 of the continuation of the first light guide 1808. The fused component longitudinal protection structure 1810 can be configured to provide structural protection to the distal tip 1814 of the proximal portion 1804 of the first light guide 1802 and the proximal tip 1816 of the continuation of the first light guide 1808 in the presence of pressure waves. The first component longitudinal protection structure 1810 can be abutted against and fused to the proximal tip 1816 of the continuation of the first light guide 1808 and the distal tip 1814 of the proximal portion 1804 of the first light guide 1802, where the first component is optically matched to the first and continuation of the first light guides, 1802 and 1808. It is appreciated that in some embodiments, the first component is not optically matched and can serve as a diverting feature.

[0102] It is appreciated that the balloons, light guides, and elongate shafts suitable for use with the protection structures can include any of those described elsewhere herein. In various embodiments, the elongate shaft defines an inflation lumen, where the inflation lumen is in fluid communication with the balloon at a distal portion of the elongate shaft.

[0103] In an example, a catheter system for imparting pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall is provided. The catheter system can include a catheter configured to advance to the vascular lesion located within or adjacent a blood vessel. The catheter can include an elongate shaft and a balloon coupled to the elongate shaft. The balloon can be configured to be filled with balloon fluid and configured to expand from a collapsed

configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to the treatment site. The elongate shaft can define a first recess within the balloon. The catheter can include a first light guide disposed along the elongate shaft and within the balloon, the first light guide configured to be placed in optical communication with a light source and a balloon fluid. A first portion of the first light guide can extend into the first recess and can include a distal tip of the first light guide. The catheter can also include a first protection structure disposed within the first recess of the elongate shaft and in contact with the first portion of the light guide, where the first protection structure includes a potting material filling the first recess, where the potting material is optically matched to the first light guide. It is appreciated that in some embodiments, the first protection structure is not optically matched.

[0104] The catheter system can include a light source that is configured to provide pulses of light to the balloon fluid, thereby initiating plasma formation in the balloon fluid, causing rapid bubble formation, and imparting pressure waves upon the vascular lesion. The first protection structure can be configured to provide structural protection to the first portion and distal tip of the first light guide in the presence of the pressure waves. In some embodiments, the first protection structure can include a potting material that fills the first recess and includes a potting material outer surface that is flush with an outer surface of the elongate shaft. In some embodiments, the first protection structure can include a potting material that partially fills the first recess and includes a potting material outer surface that is not flush with the outer surface of the elongate shaft. In yet other embodiments, the first protection structure can include a potting material that over fills the first recess and includes a potting material outer surface that forms a dome-shaped surface that extends past an outer surface of the elongate shaft, and in some embodiments the dome-shaped surface of the potting material can act as a focusing element.

[0105] The various protection structures described herein can include, but are not to be limited to, end cap protection structures, sleeve protection structures, potting material protection structures, and fused component protection structures. The protection structures can be configured to protect a portion of a light guide that can include a distal tip, a longitudinal portion, or both. In one embodiment for protection of a distal tip, an end cap protection structure can include an annular cylinder shape about a distal tip of a light guide. In one embodiment, the distal tip of the first light guide can be flush with a distal tip of the end cap and not obstructed by the end cap. In other embodiments, the distal tip of the first light guide can be flush with a distal tip of the end cap and covered by an optically matched material. In another embodiment for protection of a distal tip, a protection structures that includes potting material can be used to cover the entire outer surface of a distal tip. In various embodiments, the potting material can fill the recess so that an outer surface of the potting material is continuous with an outer surface of the elongate shaft. In some embodiments, a potting material can completely surround a distal tip of the first light guide. In other embodiments, a potting material can partially surround a distal tip of the first light guide. In yet another embodiment for protecting a distal tip, the protection structures can include fused components, where

the fused component can include a solid cylindrical shape. In some embodiments, the fused component material can be harder and more durable than the material of the light guides herein. In some embodiments, the fused component can include a material such as a glass, sapphire, diamond, and the like.

[0106] Various examples of protection structures suitable for protecting a longitudinal portion of a light guide disposed in a recess can include a sleeve, a potting material, and a fused component. In one embodiment for protection of longitudinal portion of a light guide, a sleeve protection structure can include an annular cylinder shape disposed about a longitudinal portion of a light guide. In various embodiments, the potting material can fill the recess so that an outer surface of the potting material is continuous with an outer surface of the elongate shaft. In some embodiments, a potting material can completely surround a longitudinal portion of the first light guide. In other embodiments, a potting material can partially surround a longitudinal portion of the first light guide. In some embodiments, the potting material can fill a recess and include a potting material outer surface that is flush with the outer surface of the elongate shaft. In some embodiments, the potting material can partially fill a recess and include a potting material outer surface that is not flush with the outer surface of the elongate shaft. In yet other embodiments, the potting material can over fill a recess and include a potting material outer surface that forms a dome-shaped surface that extends past an outer surface of the elongate shaft, and in some embodiments the dome-shaped surface can act as a focusing element. In yet another embodiment for protecting a longitudinal portion, the protection structures can include fused components, where the fused component can include a solid cylindrical shape disposed between two light guides. In some embodiments, the fused component material can be harder and more durable than the material of the light guides herein. In some embodiments, the fused component can include a material such as a glass, sapphire, diamond, and the like. In some examples, the first longitudinal protection structure is completely within the first longitudinal recess. In some examples, the first longitudinal protection structure protrudes from the first longitudinal recess.

[0107] The light guides including protection structures as described can be used in various methods for generating pressure waves to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. In one embodiment, the method can include advancing a catheter to a vascular lesion within the blood vessel, where the catheter includes an elongate shaft, a balloon coupled to the elongate shaft, and at least a first light guide disposed along the elongate shaft within the balloon. The first light guide can be configured to be placed in optical communication with a light source and a balloon fluid. A first portion of the first light guide can be disposed within a first recess defined by the elongate shaft, where the first portion of the first light guide can be in contact with a first light guide protection structure. The method can include expanding the balloon from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a vascular lesion. The method can include, after expanding the balloon, activating a light source in optical communication with the first light guide to direct light from within the first light guide to initiate plasma formation in the

balloon fluid and to cause rapid bubble formation, thereby imparting pressure waves upon the vascular lesion, wherein the first light guide protection structure is configured to provide structural protection to the distal tip in the presence of the pressure waves. In some embodiments, the method can include catheters where the first portion includes a distal tip of the first light guide, and where the first protection structure can be configured to provide structural protection to the distal tip of the first light guide. In various embodiments, the method can include, after activating the light source, further expanding the balloon from the first expanded configuration to a second further expanded configuration.

Focusing Element Configurations (FIGS. 19-21)

[0108] The catheter systems herein can include various focusing elements to direct light from within a light guide to a location away from the light guide. Light directed to a location away from the light guide can initiate a plasma formation within a balloon fluid at the location. The catheter systems including focusing elements are designed to impart pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall. Beneficial therapy effects may be enjoyed by initiating the plasma formation at a location away from the light guide instead of immediately adjacent to the light guide. The light guide may be less likely to be damaged by the plasma initiation event, the resulting pressure, or the resulting bubble dynamics if the plasma formation location is at a location away from the light guide rather than immediately adjacent to the light guide.

[0109] As discussed elsewhere herein, the catheter systems can include a catheter that can be configured to advance to the vascular lesion located within or adjacent a blood vessel. The catheter can include an elongate shaft and a balloon coupled to the elongate shaft. The balloon can include a balloon wall and it can be configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a vascular lesion. The catheters herein can include a first light guide disposed along the elongate shaft and within the balloon, where the first light guide can be configured to be placed in optical communication with a light source and a balloon fluid. It is appreciated that the first light guide can include an optical fiber and the light source can include a laser, both of which are described in more detail elsewhere herein.

[0110] The first light guide can include at least a first focusing element located at a distal portion of the first light guide within a balloon and in optical communication with the light source. Referring now to FIG. 19, a schematic side-view of a catheter 1900, with a partial longitudinal cross-sectional view of a balloon, is shown in accordance with various embodiments herein. Catheter 1900 includes an elongate shaft 102, a balloon 122, and a first light guide 1902 disposed along the elongate shaft 102 and within the balloon 122. The first light guide 1902 can be configured to be in optical communication with a light source at its proximal end and with a balloon fluid at its distal end. In some embodiments, the distal portion of the first light guide can include a distal tip of the first light guide. In other embodiments, the distal portion of the first light guide can include a longitudinal portion thereof. The elongate shaft 102 of catheter 1900 can define an inflation lumen (not shown),

where the inflation lumen can be in fluid communication with the balloon 122 at a distal portion of the elongate shaft 102.

[0111] In the configuration shown in FIG. 19, the first light guide 1902 includes a distal tip 1904, and a first focusing element 1906 disposed at the distal tip 1904. The first light guide 1902 can include a longitudinal axis 1908. The first focusing element 1906 can be configured to direct light 1910 from within the first light guide 1902 to a first location 1912 at a first distance 1914 away from the distal tip 1904 of the first light guide 1902. Light directed to the first location 1912 can initiate plasma formation in the balloon fluid away from the distal tip 1904 and to cause rapid bubble formation, thereby imparting pressure waves upon the vascular lesion. The first location 1912 can be spaced away from the distal tip 1904 and centered on a longitudinal axis 1908 of the first light guide 1902. The formation of plasma and bubble 1916 within the balloon fluid can originate at the first location 1912 at a first distance 1914 spaced away from the distal tip 1904 and centered on a longitudinal axis 1908 of the first light guide 1902. The first light guide 1902 can also include a first diverting feature (not shown) in optical communication with the focusing element and located at a distal portion of the first light guide, where the diverting feature is configured to direct light from within the first light guide toward a first focusing element and toward the balloon wall away from, or off of, the longitudinal axis.

[0112] In some embodiments, the first light guide 1902 can include a second focusing element located at a distal portion of the first light guide, where the second focusing element can be configured to direct light from within the first light guide to a second location at a second distance away from the distal portion of the first light guide to initiate plasma formation in the balloon fluid away from the distal portion and to cause rapid bubble formation, thereby imparting pressure waves upon the vascular lesion. The second focusing element may also be located on the distal tip along with the first focusing element 1906, such as the embodiment described with respect to FIG. 20. The second focusing element may also be located along a longitudinal surface of the light guide, similar to one of the focusing elements shown in the embodiment of FIG. 21.

[0113] In addition or alternatively, in other embodiments, catheter 1900 can include a second light guide coupled to the elongate shaft. The second light guide can be in optical communication with a light source and a balloon inflation fluid, where the second light guide can be in optical communication with the light source and the balloon inflation fluid. The second light guide can include a focusing element of the second light guide located at a distal portion of the second light guide and in optical communication with the light source. The focusing element of the second light guide can be configured to direct light from within the second light guide to a second location at a second distance from the distal portion of the second light guide to initiate plasma formation in the balloon fluid away from the distal portion and to cause rapid bubble formation, thereby imparting pressure waves upon the vascular lesion. It is appreciated that multiple light guides, each having multiple focusing elements, can be used in the catheter systems herein.

[0114] The focusing elements herein can direct light from within a light guide to one or more locations at a distance of at least 1 micrometers (μm) and at most 5 millimeters (mm) away from the distal tip of the first light guide. In an

embodiment, the focusing elements herein can direct light from within a light guide to one or more locations at a distance of at least 10 μm and at most 5 mm away from the distal tip of the first light guide. In some embodiments, the focusing elements herein can direct light from within a light guide to a distance of greater than or equal to 1 μm , 2 μm , 3 μm , 4 μm , 5 μm , 6 μm , 7 μm , 8 μm , 9 μm , 10 μm , 20 μm , 30 μm , 40 μm , 50 μm , 60 μm , 70 μm , 80 μm , 90 μm , 100 μm , 200 μm , 300 μm , 400 μm , 500 μm , 600 μm , 700 μm , 800 μm , 900 μm , 1 mm, 2 mm, 3 mm, 4 mm, or 5 mm, away from the distal tip of the first light guide, or can be an amount falling within a range between any of the foregoing. In some embodiments, the focusing elements herein can direct light from within a light guide to one or more locations at a distance greater than 1 mm.

[0115] The light guides suitable for use in the catheter systems herein can include multiple focusing elements in a distal portion, as will be discussed in reference to FIGS. 20 and 21. In the configuration in FIG. 20, a longitudinal cross-sectional view of light guide 2000 is shown in accordance with various embodiments herein. The light guide 2000 can be disposed along an elongate shaft of a catheter and placed in optical communication with a light source and a balloon fluid. The light guide 2000 can include a distal portion, where the distal portion can be a distal tip 2002. Light guide 2000 can further include a first focusing element 2004 and a second focusing element 2006. In the example of FIG. 20, both the first focusing element 2004 and the second focusing element 2006 are located at the distal tip 2002 of the light guide 2000. The first focusing element 2004 can be configured to direct light 2010 from within the light guide 2000 to a first location 2016 at a first distance 2012 spaced away from the longitudinal axis 2008 of the light guide 2000 to initiate plasma formation in the balloon fluid at the first location 2016. Formation of plasma at the first location 2016 can cause rapid bubble formation of bubble 2020, thereby imparting pressure waves upon the vascular lesion. The second focusing element 2006 can be configured to direct light 2010 from within the light guide 2000 to a second location 2018 at a second distance 2014 spaced away from the longitudinal axis 2008 of the light guide 2000 to initiate plasma formation in the balloon fluid away at the second location 2018. Formation of plasma at the second location 2018 can cause rapid bubble 2022 formation, thereby imparting pressure waves upon the vascular lesion. In the embodiment shown in FIG. 20, the first distance 2012 is equal to or approximately equal to the second distance 2014, the first location 2016 is off-axis in one direction, and the second location 2018 is off-axis in a different direction. In some embodiment, the first distance 2012 is not equal to the second distance 2014.

[0116] In the configuration in FIG. 21, a longitudinal cross-sectional view of light guide 2100 is shown in accordance with various embodiments herein. The light guide 2100 can be disposed along an elongate shaft of a catheter and placed in optical communication with a light source and a balloon fluid. The light guide 2100 can include a distal portion, such as a longitudinal distal portion 2102. The light guide 2100 can include a first diverting feature 2124 located at a distal portion of the first light guide and configured to direct light from within the first light guide toward a first focusing element 2104. In some examples, the first focusing element 2104 can be configured to direct light from within the light guide 2100 to a location in the balloon fluid away

from a longitudinal axis 2108 of the first light guide 2100. In some examples, the light guide 2100 can include a second diverting feature 2126, where the second diverting feature is located at a distal portion of the first light guide and configured to direct light from within the first light guide toward a second focusing element 2106. In some examples, the second focusing element 2106 can be configured to direct light from within the light guide 2100 to a location in the balloon fluid away from a longitudinal axis 2108 of the elongate shaft.

[0117] The first focusing element 2104 can be configured to direct light 2110 from within the light guide 2100 to a first location 2116 at a first distance 2112 from the first light guide 2100, which is spaced away from the longitudinal axis 2108 of the light guide 2100 to initiate plasma formation in the balloon fluid at the first location 2116 and to cause rapid bubble 2120 formation, thereby imparting pressure waves upon the vascular lesion. The second focusing element 2106 can be configured to direct light 2110 from within the light guide 2100 to a second location 2118 at a second distance 2114 from the first light guide 2100, which is spaced away from the longitudinal axis 2108 of the light guide 2100 to initiate plasma formation in the balloon fluid away at the second location 2118 and to cause rapid bubble 2122 formation, thereby imparting pressure waves upon the vascular lesion. In the embodiment shown in FIG. 21, the first distance 2112 and second distance 2114 are the same. In some embodiments, the first distance 2112 and second distance 2114 can be different. It is appreciated that the diverting features can be tailored to increase or decrease the distance that the light from within the light guide is focused at a location spaced away from the longitudinal axis.

[0118] Diverting features suitable for use with the focusing elements herein such as the embodiment of FIG. 21, can include, but are not to be limited to, any feature of the system herein that diverts light from the light guide away from its axial path toward a side surface portion of the light guide. Examples include a reflector, a refracting element, and a fiber diffuser, as will be discussed in more detail below. It is appreciated that the light guides used in the catheter systems herein can include a second, third, fourth, fifth, or sixth focusing element. In some embodiments, the light guides herein can include more than six focusing elements. The focusing elements suitable for use herein can include one or more structures such as a convex lens, a convex mirror, or a gradient-index (GRIN) lens. In some examples, the focusing elements utilized in the light guides herein can be a GRIN lens. In other examples, the focusing elements utilized in the light guides herein can include convex lenses. In yet other embodiments, the focusing elements utilized in the light guides herein can include convex mirrors. In various embodiments, a GRIN lens, convex lens, or convex mirror can be adhered to or fused to a side surface portion or distal tip of the light guides herein with an optically matched adhesive. Adhesives suitable for use with the focusing elements herein can include optically matched or optically mismatched adhesives.

[0119] The focusing elements can be located at a distal tip of the light guides herein or disposed along the longitudinal axis of the light guides. The focusing elements described herein can be present in a distal tip of the light guides or along one or more locations along a longitudinal portion of the light guides.

[0120] The light guides including focusing elements as described can be used in various methods for generating pressure waves to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. In one embodiment, the method can include advancing a catheter to a vascular lesion within the blood vessel, where the catheter includes an elongate shaft, a balloon coupled to the elongate shaft, and at least a first light guide disposed along the elongate shaft within the balloon. The first light guide can be configured to be placed in optical communication with a light source and a balloon fluid. The method can include expanding the balloon from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a vascular lesion. The method can further include, after expanding the balloon, activating a light source in optical communication with the first light guide to direct light from within the first light guide to a first location at a first distance away from the distal portion of the first light guide to initiate plasma formation in the balloon fluid away from the distal portion and to cause rapid bubble formation, thereby imparting pressure waves upon the vascular lesion. The first location is spaced away from the distal tip and is centered on a longitudinal axis of the first light guide. In various embodiments, the first location can be spaced away from the distal end of the light guide such that it is off-axis from the longitudinal axis of the light guide, or away from the longitudinal axis of the light guide. The method can further include, after activating the light source, further expanding the balloon from the first expanded configuration to a second further expanded configuration.

Fortified Balloon Inflation Fluids (FIGS. 22 and 23)

[0121] The balloons herein can be inflated with a fortified balloon inflation fluid that is configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid when compared to a base inflation fluid. The fortified balloon inflation fluid can be used in the catheter systems embodied herein. Briefly, the fortified balloon inflation fluid can be used in a catheter system for imparting pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall. The catheter systems can include a catheter that can be configured to advance to the vascular lesion located within or adjacent a blood vessel. The catheter can include an elongate shaft, where a balloon can be coupled to the elongate shaft, and where the balloon includes a balloon wall. The elongate shaft can define an inflation lumen, where the inflation lumen can be in fluid communication with the balloon at a distal portion of the elongate shaft and in fluid communication with a fluid source at a proximal end of the elongate shaft. The balloon can be configured to expand from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a vascular lesion. The balloon can also be configured to expand to a second further expanded configuration. The catheter suitable for use with the fortified balloon inflation fluid can assume many configurations as discussed elsewhere herein.

[0122] The fortified balloon inflation fluids described herein can include a base inflation fluid and a fortification component. The fortification component can reduce a

threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid. In some embodiments, the fortification component can include carbon or iron. In some embodiments the fortification element can include iron dextran. In other embodiments, the fortification component can include carbon. In yet other embodiments, the fortification component can include nanoparticles. Various examples of fortification components suitable for use herein include, but are not to be limited to, iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, including, but not to be limited to single walled carbon nanotubes or double walled carbon nanotubes or mixtures thereof, gold-coated carbon nanotubes, or copper-coated carbon nanotubes. The fortification component can modify various physical parameters of the fortified balloon inflation fluid, such as, but not limited to, viscosity, density, or surface tension. The fortification component can be configured to increase or decrease one or more the viscosity, density, or surface tension of the fortified balloon inflation fluid compared to the base inflation fluid.

[0123] The base inflation fluids herein can include those having a mixture of saline and contrast medium. The ratios of the saline and contrast medium can be tailored for treatment at vascular lesion within a vessel wall. In some embodiment, the saline and contrast medium can be present within the base inflation fluid in a ratio of saline to contrast medium of 25:75 volume percent to 75:25 volume percent. In some examples, the ratio of saline to contrast medium within the base inflation fluid can be 25:75 volume percent. In other examples, the ratio of saline to contrast medium within the base inflation fluid can be 50:50 volume percent. In yet other examples, the ratio of saline to contrast medium within the base inflation fluid can be 75:25 volume percent.

[0124] Fortified balloon inflation fluids having iron dextran as the fortification component can include a concentration of iron dextran from at least 0.0001 (millimole per liter) mmol/L to 1.0 mmol/L. In some embodiments, the concentration of iron dextran can be greater than or equal to 0.0001 mmol/L, 0.0002 mmol/L, 0.0003 mmol/L, 0.0004 mmol/L, 0.0005 mmol/L, 0.0006 mmol/L, 0.0007 mmol/L, 0.0008 mmol/L, 0.0009 mmol/L, 0.001 mmol/L, 0.002 mmol/L, 0.003 mmol/L, 0.004 mmol/L, 0.005 mmol/L, 0.006 mmol/L, 0.007 mmol/L, 0.008 mmol/L, 0.009 mmol/L, 0.01 mmol/L, 0.02 mmol/L, 0.03 mmol/L, 0.04 mmol/L, 0.05 mmol/L, 0.06 mmol/L, 0.07 mmol/L, 0.08 mmol/L, 0.09 mmol/L, 0.1 mmol/L, 0.2 mmol/L, 0.3 mmol/L, 0.4 mmol/L, 0.5 mmol/L, 0.6 mmol/L, 0.7 mmol/L, 0.8 mmol/L, 0.9 mmol/L, or 1.0 mmol/L or can be an amount falling within a range including any of the foregoing.

[0125] Fortified balloon inflation fluids having nanoparticles as the fortification component can include a concentration of nanoparticles from at least 0.01 weight per volume percent (w/v %) to 15 w/v %. In some embodiments, the concentration of nanoparticles present in the fortified balloon inflation fluids can be greater than or equal to 0.01 w/v %, 0.02 w/v %, 0.03 w/v %, 0.04 w/v %, 0.05 w/v %, 0.06 w/v %, 0.07 w/v %, 0.08 w/v %, 0.09 w/v %, 0.10 w/v %, 0.2 w/v %, 0.3 w/v %, 0.4 w/v %, 0.5 w/v %, 0.6 w/v %, 0.7 w/v %, 0.8 w/v %, 0.9 w/v %, 1 w/v %, 2 w/v %, 3 w/v %, 4 w/v %, 5 w/v %, 6 w/v %, 7 w/v %, 8 w/v %, 9 w/v %, 10 w/v %, 11 w/v %, 12 w/v %, 13 w/v %, 14 w/v %, or 15 w/v %, or can be an amount falling within a range including any of the foregoing.

[0126] The fortified balloon inflation fluid can be used in catheter systems herein that include a first light guide disposed along the elongate shaft and within the balloon, where the first light guide can be configured to be placed in optical communication with a light source and the fortified balloon inflation fluid. The fortified balloon inflation fluid can be used in catheter systems herein that include a second light guide, a third light guide, a fourth light guide, or more than four light guides. The light guides suitable for use with the fortified balloon inflation fluid can include any of the light guides configured as described elsewhere herein. The light source used with the fortified balloon inflation fluid can be configured to provide sub-millisecond pulses of a light from the light source to at least the first light guide, thereby initiating plasma formation in the fortified balloon inflation fluid, causing rapid bubble formation, and imparting pressure waves upon the vascular lesion.

[0127] The fortification component can be included as a coating on one or more surfaces of the catheter systems herein, where it can be solvated by a base inflation fluid prior to use in treatment at a treatment site. Referring now to FIG. 22, a longitudinal cross section of a catheter 2200 is shown in accordance with various embodiments herein. The catheter 2200 can be used in a catheter system for imparting pressure to induce fractures in a vascular lesion within or adjacent a blood vessel wall. The catheter 2200 can be configured to advance to the vascular lesion located within or adjacent a blood vessel. The catheter 2200 can include an elongate shaft 102 and a balloon 122 coupled to the elongate shaft 102. The balloon 122 can include a balloon wall. The catheter 2200 can include a fortified balloon inflation fluid 2201. In various embodiments, the fortified balloon inflation fluid 2201 can include a fortification component 2202 and a base inflation fluid 2203 (illustrated by a plurality of “X”s). The fortification component 2202 can be in the form of a fortification component coating disposed on an inside surface of the balloon 122 and in fluid communication with a base inflation fluid 2203. The fortification component 2202 can include a fortification component coating that comprises carbon and/or iron, as discussed herein. As used herein, the terms “fortification component” and “fortification component coating” can be used interchangeably.

[0128] The balloon 122 of catheter 2200 can be configured to expand from a collapsed configuration suitable for advancing the catheter 2200 through a patient’s vasculature to a first expanded configuration suitable for anchoring the catheter 2200 in position relative to a vascular lesion. The balloon 122 of catheter 2200 can be inflated with a base inflation fluid, where the base inflation fluid 2203 is configured to solvate the fortification component coating on the inside surface of the balloon to form the fortified balloon inflation fluid 2201. The fortification component 2202 of the fortified balloon inflation fluid 2201 can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid 2201 compared to the base inflation fluid 2203.

[0129] The catheter 2200 can include a first light guide 2204 disposed along the elongate shaft 102 and within the balloon 122, the first light guide 2204 can be configured to be placed in optical communication with a light source and the fortified balloon inflation fluid. The catheter 2200 can also include a second light guide coupled to the elongate shaft, where the second light guide can be in optical communication with the light source and the fortified balloon

inflation fluid. The light source can be configured to provide sub-millisecond pulses of light from the light source to at least the first light guide 2204, and if present any additional light guides, thereby initiating plasma formation in the fortified balloon inflation fluid, causing rapid bubble formation, and imparting pressure waves upon the vascular lesion. The first light guide 2204 can be an optical fiber and the light source can be a laser, both of which are described in more detail elsewhere herein.

[0130] The catheters herein can also include a fortification component within a fortification component coating disposed along the elongate shaft. Referring now to FIG. 23, a longitudinal cross section of a catheter 2300 is shown in accordance with various embodiments herein. Catheter 2300 is similar to catheter 2200 of FIG. 22, in that it has similar components including an elongate shaft 102 and a balloon 122 coupled to the elongate shaft 102. The balloon 122 can include a balloon wall. The catheter 2300 can include a fortification component 2302 in the form of a fortification component coating 2302 that is disposed on a surface of the elongate shaft 102 and in fluid communication with the base inflation fluid 2301. The fortification component coating 2302 can include a fortification component that comprises carbon and/or iron, as discussed herein.

[0131] The balloon 122 of catheter 2300 can be configured to expand from a collapsed configuration suitable for advancing the catheter 2300 through a patient’s vasculature to a first expanded configuration suitable for anchoring the catheter 2300 in position relative to a vascular lesion. The balloon 122 of catheter 2300 can be inflated with a base inflation fluid 2303 (illustrated by a plurality of “X”s), where the base inflation fluid 2303 is configured to solvate the fortification component 2302 disposed on a surface of the elongate shaft 102 to form a fortified balloon inflation fluid 2301. The fortification component 2302 of the fortified balloon inflation fluid 2301 can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid 2302 compared to the base inflation fluid 2303.

[0132] The catheter 2300 can include a first light guide 2304 disposed along the elongate shaft 102 and within the balloon 122, the first light guide 2304 can be configured to be placed in optical communication with a light source and the fortified balloon inflation fluid. The catheter 2300 can also include a second light guide coupled to the elongate shaft, where the second light guide can be in optical communication with the light source and the fortified balloon inflation fluid 2301. The light source can be configured to provide sub-millisecond pulses of light from the light source to at least the first light guide 2304, and if present any additional light guides, thereby initiating plasma formation in the fortified balloon inflation fluid 2301, causing rapid bubble formation, and imparting pressure waves upon the vascular lesion. The first light guide 2304 can be an optical fiber and the light source can be a laser, both of which are described in more detail elsewhere herein.

[0133] The fortified balloon inflation fluids described herein can be used in various ways for generating pressure waves to induce fractures in a vascular lesion within or adjacent a vessel wall of a blood vessel. In one embodiment, the method can include advancing a catheter to a vascular lesion within the blood vessel, where the catheter includes an elongate shaft, a balloon coupled to the elongate shaft, and at least a first light guide disposed along the elongate

shaft within the balloon. The method can include expanding the balloon from a collapsed configuration suitable for advancing the catheter through a patient's vasculature to a first expanded configuration suitable for anchoring the catheter in position relative to a vascular lesion. The step of expanding the balloon can include expanding the balloon with a fortified balloon inflation fluid including a base inflation fluid and a fortification component, where the fortification component can be configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid. In various embodiments, the method of expanding the balloon can include providing the fortification component as a coating disposed on an inside surface of a balloon wall or on the elongate shaft, where the fortification component coating is in fluid communication with the base inflation fluid, and providing the base inflation fluid to the balloon, where the base inflation fluid solvates the fortification component coating to form the fortified balloon inflation fluid.

[0134] The fortification component can include carbon or iron, where the fortification component can include, but is not to be limited to, iron dextran or nanoparticles. Some exemplary nanoparticles include iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, including, but not to be limited to single walled carbon nanotubes or double walled carbon nanotubes or mixtures thereof, gold-coated carbon nanotubes, or copper-coated carbon nanotubes. In various embodiments, after expanding the balloon, the method can include activating a light source in optical communication with the light guide and the fortified balloon inflation fluid to provide sub-millisecond pulses of light from the light source to the fortified balloon inflation fluid, thereby initiating plasma formation in a fortified balloon inflation fluid and causing rapid bubble formation, and imparting pressure waves upon the vascular lesion. In some embodiments, after activating the light source, the method can include further expanding the balloon from the first expanded configuration to a second further expanded configuration. In other embodiments, after activating the light source, the method can include further expanding the balloon from the first expanded configuration to a second further expanded configuration.

[0135] The light sources herein can be configured to generate sub-millisecond pulses of light to be delivered to the treatment site at a frequency of from at least 1 hertz (Hz) to 5000 Hz. In some embodiments, the light sources herein can be configured to generate sub-millisecond pulses of light to be delivered to the treatment site at a frequency from at least 30 Hz to 1000 Hz. In other embodiments, the light sources herein can be configured to generate the sub-millisecond pulses of light to be delivered to the treatment site at a frequency from at least 10 Hz to 100 Hz. In yet other embodiments, the light sources herein can be configured to generate sub-millisecond pulses of light to be delivered to the treatment site at a frequency that can be greater than or equal to 1 Hz, 2 Hz, 3 Hz, 4 Hz, 5 Hz, 6 Hz, 7 Hz, 8 Hz, or 9 Hz, 10 Hz, 20 Hz, 30 Hz, 40 Hz, 50 Hz, 60 Hz, 70 Hz, 80 Hz, 90 Hz, 100 Hz, 200 Hz, 300 Hz, 400 Hz, 500 Hz, 600 Hz, 700 Hz, 800 Hz, 900 Hz, 1000 Hz, 1250 Hz, 1500 Hz, 1750 Hz, 2000 Hz, 2250 Hz, 2500 Hz, 2750 Hz, 3000 Hz, 3250 Hz, 3500 Hz, 3750 Hz, 4000 Hz, 4250

Hz, 4500 Hz, 4750 Hz, or 5000 Hz or can be an amount falling within a range between any of the foregoing.

[0136] Balloons

[0137] The balloons suitable for use in the catheter systems herein include those that can be passed through the vasculature of a patient when in a collapsed configuration. In some embodiments, the balloons herein are made from silicone. In other embodiments, the balloons herein are made from polydimethylsiloxane (PDMS), polyurethane, polymers such as PEBAX™ material available from Arkema, which has a location at King of Prussia, Pa., USA, nylon, and the like. In some embodiments, the balloons can include those having diameters ranging from 1 millimeter (mm) to 25 mm in diameter. In some embodiments, the balloons can include those having diameters ranging from at least 1.5 mm to 12 mm in diameter. In some embodiments, the balloons can include those having diameters ranging from at least 1 mm to 5 mm in diameter. In some embodiments, the diameter can be greater than or equal to 0.5 mm, 1.0 mm, 1.5 mm, 2.0 mm, 2.5 mm, 3.0 mm, 3.5 mm, 4.0 mm, 4.5 mm, 5.0 mm, 5.5 mm, 6.0 mm, 6.5 mm, 7.0 mm, 7.5 mm, 8.0 mm, 8.5 mm, 9.0 mm, 9.5 mm, 10.0 mm, 10.5 mm, 11.0 mm, 11.5 mm, 12.0 mm, 12.5 mm, 13.0 mm, 13.5 mm, 14.0 mm, 14.5 mm, 15.0 mm, 15.5 mm, 16.0 mm, 16.5 mm, 17.0 mm, 17.5 mm, 18.0 mm, 18.5 mm, 19.0 mm, 19.5 mm, or 20.0 mm, or can be an amount falling within a range between any of the foregoing.

[0138] In some embodiments, the balloons herein can include those having a length ranging from at least 5 mm to 300 mm in length. In some embodiments, the balloons herein can include those having a length ranging from at least 8 mm to 200 mm in length. In some embodiments, the length of the balloon can be greater than or equal to 5 mm, 10 mm, 20 mm, 30 mm, 40 mm, 50 mm, 60 mm, 70 mm, 80 mm, 90 mm, 100 mm, 110 mm, 120 mm, 130 mm, 140 mm, 150 mm, 160 mm, 170 mm, 180 mm, 190 mm, 200 mm, 210 mm, 220 mm, 230 mm, 240 mm, 250 mm, 260 mm, 270 mm, 280 mm, 290 mm, or 300 mm, or can be an amount falling within a range between any of the foregoing.

[0139] The balloons herein can be inflated to inflation pressures from 1 atmosphere (atm) to 70 atm. In some embodiments, the balloons herein can be inflated to inflation pressures of from at least 20 atm to 70 atm. In some embodiments, the balloons herein can be inflated to inflation pressures of from at least 6 atm to 20 atm. In some embodiments, the balloons herein can be inflated to inflation pressures of from at least 3 atm to 20 atm. In some embodiments, the balloons herein can be inflated to inflation pressures of from at least 2 atm to 10 atm. In some embodiments, the balloons herein can be inflated to inflation pressures that can be greater than or equal to 1 atm, 2 atm, 3 atm, 4 atm, 5 atm, 6 atm, 7 atm, 8 atm, 9 atm, 10 atm, 15 atm, 20 atm, 25 atm, 30 atm, 35 atm, 40 atm, 45 atm, 50 atm, 55 atm, 60 atm, 65 atm, or 70 atm, or can be an amount falling within a range between any of the foregoing.

[0140] The balloons herein can include those having various shapes, including, but not to be limited to, a conical shape, a cylindrical shape, a square shape, a rectangular shape, a spherical shape, a conical/square shape, a conical/spherical shape, an extended spherical shape, an oval shape, a tapered shape, a bone shape, a stepped diameter shape, an offset shape, or a conical offset shape. In some embodiments, the balloons herein can include a drug eluting coating or a drug eluting stent structure. The drug elution coating or

drug eluting stent can include one or more therapeutic agents including anti-inflammatory agents, anti-neoplastic agents, anti-angiogenic agents, and the like.

Balloon Fluids

[0141] Exemplary balloon fluids suitable for use herein can include, but are not to be limited to one or more of water, saline, contrast medium, fluorocarbons, perfluorocarbons, gases, such as carbon dioxide, and the like. In some embodiments, the balloon fluids described can be used as base inflation fluids, discussed elsewhere herein. In some embodiments, the balloon inflation fluids include a mixture of saline to contrast medium in a volume ratio of 50:50. In some embodiments, the balloon fluids include a mixture of saline to contrast medium in a volume ratio of 25:75. In some embodiments, the balloon fluids include a mixture of saline to contrast medium in a volume ratio of 75:25. The balloon fluids suitable for use herein can be tailored on the basis of composition, viscosity, and the like in order to manipulate the rate of travel of the pressure waves therein. The balloon fluids suitable for use herein are biocompatible. A volume of balloon fluid can be tailored by the chosen light source and the type of balloon fluid used.

[0142] In some embodiments, the contrast agents used in the contrast media herein can include, but are not to be limited to, iodine-based contrast agents, such as ionic or non-ionic iodine-based contrast agents. Some non-limiting examples of ionic iodine-based contrast agents include diatrizoate, metrizoate, iohalamate, and ioxaglate. Some non-limiting examples of non-ionic iodine-based contrast agents include iopamidol, iohexol, ioxilan, iopromide, iodixanol, and ioversol. In other embodiments, non-iodine based contrast agents can be used. Suitable non-iodine containing contrast agents can include gadolinium (III)-based contrast agents. Suitable fluorocarbon and perfluorocarbon agents can include, but are not to be limited to, agents such as the perfluorocarbon dodecafluoropentane (DDFP, C₅F₁₂).

[0143] The balloon fluids herein can include those that include absorptive agents that can selectively absorb light in the ultraviolet (e.g., at least 10 nanometers (nm) to 400 nm), visible region (e.g., at least 400 nm to 780 nm), and near-infrared region of the electromagnetic spectrum (e.g., at least 780 nm to 2.5 μ m), or in the far-infrared region of the electromagnetic spectrum of at least 2.5 micrometers to 10 micrometers (μ m). Suitable absorptive agents can include those with absorption maxima along the spectrum from at least 10 nm to 2.5 μ m. In various embodiments, the absorptive agent can be those that have an absorption maximum matched with the emission maximum of the laser used in the catheter system. By way of non-limiting examples, various lasers described herein can include neodymium:yttrium-aluminum-garnet (Nd:YAG-emission maximum=1064 nm) lasers, holmium:YAG (Ho:YAG-emission maximum=2.1 μ m) lasers, or erbium:YAG (Er:YAG-emission maximum=2.94 μ m). In some embodiments, the absorptive agents used herein can be water soluble. In other embodiments, the absorptive agents used herein are not water soluble. In some embodiments, the absorptive agents used in the balloon fluids herein can be tailored to match the peak emission of the light source. Various light sources having emission wavelengths of at least 10 nanometers to greater than 1 millimeter are discussed elsewhere herein.

[0144] In some embodiments, introduction of the balloon fluid causes the expansion of the balloon from a collapsed

configuration to a first expanded configuration and from a first expanded configuration to a second further expanded configuration. In addition or alternatively, the expansion of the balloon can be accomplished using a shape-memory material or other means.

Light Guides (FIGS. 24-29)

[0145] The light guides herein can include an optical fiber or flexible light pipe. The light guides herein can be thin and flexible and can allow light signals to be sent with very little loss of strength. The light guides herein can include a core surrounded by a cladding about its circumference. In some embodiments, the core can be a cylindrical core or a partially cylindrical core. The core and cladding of the light guides can be formed from one or more materials, including but not limited to one or more types of glass, silica, or one or more polymers. The light guides may also include a protective coating, such as a polymer. It is appreciated that the index of refraction of the core will be greater than the index of refraction of the cladding.

[0146] Each light guide can guide light along its length to a distal portion having at least one optical window. The light guides can create a light path as portion of an optical network including a light source. The light path within the optical network allows light to travel from one part of the network to another. Both the optical fiber and/or the flexible light pipe can provide a light path within the optical networks herein.

[0147] The light guides herein can assume many configurations about the elongate shaft of the catheters described herein. In some embodiments, the light guides can run parallel to the longitudinal axis of the elongate shaft of the catheter. In some embodiments, the light guides can be disposed spirally or helically about the longitudinal axis of the elongate shaft of the catheter. In some embodiments, the light guides can be physically coupled to the elongate shaft. In other embodiments, the light guides can be disposed along the length of the outer diameter of the elongate shaft. In yet other embodiments the light guides herein can be disposed within one or more light guide lumens within the elongate shaft. Various configurations for the elongate shafts and light guide lumens will be discussed below.

[0148] The light guides herein can include various configurations at a distal portion of the light guide. Referring now to FIGS. 24-29, schematic cross-sectional views of the distal portions of various shaped light guides are shown in accordance with various embodiments herein. In FIG. 24, a schematic cross-sectional view of a light guide 2400 is shown. Light guide 2400 includes a cylindrical end shape. In some embodiments, the end of the light guide can have a tapered shape. By way of example, in FIG. 25 a schematic cross-sectional view of a light guide 2500 having a tapered end shape is shown. In some embodiments, the end of the light guide can have an angled shape. By way of example, in FIG. 26 a schematic cross-sectional view of a light guide 2600 is shown. Light guide 2600 includes an angled end shape. In some embodiments, a light guide with an angled shape can include a diverting feature. By way of example, in FIG. 27 a schematic cross-sectional view of a light guide 2700 is shown. The light guide 2700 also includes a diverting feature 2706 at the distal portion to direct the light 2704 within the light guide toward the side surface portion 2708 of the light guide. Light guide 2700 is configured such that light 2704 travels from a light source (not shown) in the

direction from the proximal portion of the light guide to the distal portion of the light guide **2700**, as indicated by the arrow. Upon contact with the diverting feature **2706**, the light **2704** is diverted, or reflected, within the light guide **2700**.

[0149] In some embodiments, a diverting feature can be included with the light guide to direct light toward a side surface portion of the distal portion of the light guide. A diverting feature can include any feature of the system herein that diverts light from the light guide away from its axial path toward a side surface portion of the light guide. Examples include a reflector, a refracting element, and a fiber diffuser. Fiber diffusers will be discussed in more detail below.

[0150] The light guides herein can also include one or more focusing elements for directing the origin of a pressure wave away from the distal tip of the light guides. By way of example, in FIG. **28** a schematic cross-sectional view of a light guide **2800** is shown. The light guide **2800** includes a first convex surface **2802**. The first convex surface **2802** is configured to direct light away from the distal tip of the light guide **2800** to generate a pressure wave having an origin point away from the surface of the distal tip. In some embodiments, the light guides in accordance with the embodiments therein can be configured to include multiple convex surfaces. By way of example, in FIG. **29** a schematic cross-sectional view of a light guide **2900** is shown. The light guide **2900** includes a first convex surface **2902** and a second convex surface **2904**. The first convex surface **2902** and second convex surface **2904** can be configured to direct light away from the distal tip of the light guide to generate a plurality of pressure waves each having a plurality of origin points away from the surface of the distal tip.

[0151] In other embodiments, the light guides can form a spiral configuration about the longitudinal axis of the elongate shaft of the catheter. In some embodiments, the spiral configuration can run clockwise about the longitudinal axis of the elongate shaft of the catheter, while in other embodiments the spiral configuration can run counter-clockwise about the longitudinal axis of the elongate shaft of the catheter. In some embodiments, the light guides can form a single helix, a double helix, a triple helix, or a quadruple helix about the longitudinal axis of the elongate shaft of the catheter.

[0152] The light guides herein can come in various sizes and configurations. The light guides will have a longitudinal axis along the elongate shaft of the light guide and short axis about its circumference. In some embodiments, the light guides can have an outer diameter of about 100 μm , including the cladding and the core. In other embodiments, the light guides can include those that have an outer diameter of from 50 μm to 1000 μm including the cladding and the core. The length of the light guides can include those having a length of from 40 cm to 175 cm. In some embodiments, the length of the light guides can include those having a length of from 50-150 cm. In some embodiments, the length of the light guide can include those having a length of 40 cm, 50 cm, 60 cm, 70 cm, 80 cm, 90 cm, 100 cm, 125 cm, 150 cm, or 175 cm. It is appreciated that the light guides herein can have a usable length that can fall within a range, wherein any of the foregoing lengths can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

[0153] It is appreciated that one or more light guides herein can be adhered to the outside surface of the elongate shaft of a catheter, to create a catheter. However, in other embodiments, one or more light guides can be disposed within a lumen of a catheter. In addition, the catheter may define a lumen for a guidewire having an inner diameter of about 0.014 inch (0.356 mm). In some embodiments, the catheter can include those having an inner diameter of about 0.018 inch (0.457 mm). In yet other embodiments, the catheter can include those having an inner diameter of about 0.035 inch (0.889 mm). In some embodiments the light guides herein can be integrated with a balloon catheter. In some embodiments the light guides herein can be integrated into a guidewire. In embodiments where the light guide is integrated into a guidewire, the resulting catheter can be used independently or can be used with various other balloon catheters.

Lumens of the Elongate Shaft (FIGS. **30-41**)

[0154] The elongate shafts herein can include one or more lumens that span the length of the elongate shaft. Referring now to FIGS. **30-41**, schematic cross-sectional views of various embodiments of an elongate shaft having multiple lumens are shown in accordance with various embodiments herein. In some embodiments, the elongate shaft can define a guidewire lumen. In some embodiments, the elongate shaft defines an inflation lumen surrounding the guidewire lumen, where the inflation lumen is in fluid communication with a balloon at a distal portion of the elongate shaft. In other embodiments, the elongate shaft defines an inflation lumen disposed alongside the guidewire lumen, where the inflation lumen is in fluid communication with a balloon at a distal portion of the elongate shaft. In yet other embodiments, the elongate shaft defines at least one control lumen, at least one light guide lumen, or at least one drug therapy lumen.

[0155] In the configuration in FIG. **30**, elongate shaft **3000** includes concentrically disposed guidewire lumen **3002** and an inflation lumen **3004**. In the configuration in FIG. **31**, elongate shaft **3100** includes guidewire lumen **3102** and an inflation lumen **3104** disposed adjacent to and partially surrounding guidewire lumen **3102**. In the configuration in FIG. **32**, elongate shaft **3200** includes guidewire lumen **3202** and an inflation lumen **3204** disposed adjacent to guidewire lumen **3202**. In the configuration in FIG. **33**, elongate shaft **3300** includes guidewire lumen **3302**, inflation lumen **3304**, and a control lumen **3306**. It is appreciated that the control lumens herein can be used for many purposes, including, but not be limited to, blood flow, cooling or heating fluid flow, delivery of a diagnostic or therapeutic agent, light guides, and the like. In the configuration in FIG. **34**, elongate shaft **3400** includes guidewire lumen **3402**, inflation lumen **3404**, and two control lumens **3406** and **3408**. In the configuration in FIG. **35**, elongate shaft **3500** includes guidewire lumen **3502**, inflation lumen **3504**, and control lumen **3506**.

[0156] The light guides can be disposed within one or more light guide lumens disposed within the elongate shafts symmetrically about the circumference. In some embodiments, the lumens herein can include those that are used for blood flow, cooling or heating fluid flow, delivery of a diagnostic or therapeutic agent, and the like. In the configuration in FIG. **36**, elongate shaft **3600** includes guidewire lumen **3602**, light guide lumen **3604**, and control lumen **3606**. One or more of lumens **3602**, **3604** and **3606** can serve as an inflation lumen. In the configuration in FIG. **37**,

elongate shaft **3700** includes guidewire lumen **3702**, light guide lumen **3704**, and control lumen **3706**. Elongate shaft **3700** includes two additional lumens that can both be configured as light guide lumens, control lumens, or both a light guide lumen and control lumen. One or more of lumens **3702**, **3704** and **3706** can serve as an inflation lumen. In the configuration in FIG. **38**, elongate shaft **3800** includes guidewire lumen **3802**, light guide lumen **3804**, and control lumen **3806**. Elongate shaft **3800** includes six additional lumens that can be configured as inflation lumens, light guide lumens, control lumens, or any combination of inflation lumens, light guide lumens and control lumens.

[0157] The light guides can be disposed within one or more light guide lumens disposed within the elongate shafts asymmetrically about the circumference. In the configuration in FIG. **39**, elongate shaft **3900** includes guidewire lumen **3902**, light guide lumen **3904**, and control lumen **3906**. Elongate shaft **3900** includes one additional lumen that can be configured as a light guide lumen **3904** or a control lumen **3906**. In the configuration in FIG. **40**, elongate shaft **4000** includes guidewire lumen **4002**, light guide lumen **4004**, and control lumen **4006**. Elongate shaft **4000** includes three additional lumens that can be configured as light guide lumens, control lumens, or any combination of light guide lumens and control lumens. In the configuration in FIG. **41**, elongate shaft **4100** includes guidewire lumen **4102**, light guide lumen **4104**, and control lumen **4106**. Elongate shaft **4100** includes three additional lumens that can be configured as inflation lumens, light guide lumens, control lumens, or any combination of inflation lumens, light guide lumens, and control lumens.

[0158] It is appreciated that the lumens described in FIGS. **30-41** can assume many shapes, including, but not to be limited to, circular shape, square shape, crescent shape, triangular shape, and the like. The lumens of the elongate shafts can be symmetrically distributed in the elongate shaft, asymmetrically distributed, or concentrically distributed. It will be further appreciated that the light guide lumens herein can be coated along the longitudinal length of the elongate shaft with a reflective material capable of propagating light along the elongate shaft from a distal light source to the proximal portion of the catheter, so that the lumen itself can act as a light guide without a separate fiber optic structure.

Diverting Features

[0159] The diverting features suitable for use herein include a reflecting element, a refracting element, and a fiber diffuser. In some embodiments, the diverting feature can be a reflecting element. In some embodiments, the diverting feature can be a refracting element. In some embodiments, the diverting feature can be a fiber diffuser.

[0160] A fiber diffuser can direct light from within a light guide to exit at a side surface portion of the light guide. The fiber diffusers described herein can be created several ways. In some embodiments, the fiber diffusers can be created by micro-machining the surface of the distal portion of a light guide with a CO₂ laser. In some embodiments, a fused silica coating can be applied to the distal portion of the light guide. In other embodiments, the fiber diffuser can be formed from a glass, a polymer, or a metal coating on the distal portion of the light guide. In other embodiments, the fiber diffuser can be formed by a fiber Bragg grating on the distal portion of the light guide. In some embodiments, the fiber diffuser can include a machined portion of the light

guide, a laser-machined portion of the light guide, fiber Bragg gratings, a fused splicing, a fused splicing forming at least one internal mirror, and a splicing of two or more diffuse regions.

[0161] Suitable materials for a fiber diffuser can include, but are not be limited to, the materials of the light guide core or light guide cladding, ground glass, silver coated glass, gold coated glass, TiO₂, and other materials that will scatter and not significantly absorbed the light wavelength of interest. One method that can be used to create a uniform diffuser in a light guide, optical component, or materials is to utilize scattering centers on the order of at least 50 nanometers to 5 micrometers in size. The scattering centers can have a distribution about 200 nanometers in size.

[0162] The diverting features suitable for focusing light away from the tip of the light guides herein can include, but are not to be limited to, those having a convex surface, a gradient-index (GRIN) lens, and a mirror focus lens.

Light Sources

[0163] The light sources suitable for use herein can include various types of light sources including lasers and lamps. Suitable lasers can include short pulse lasers on the sub-millisecond timescale. In some embodiments, the light source can include lasers on the nanosecond (ns) timescale. The lasers can also include short pulse lasers on the picosecond (ps), femtosecond (fs), and microsecond (us) timescales. It is appreciated that there are many combinations of laser wavelengths, pulse widths and energy levels that can be employed to achieve plasma in the balloon fluid of the catheters described herein. In various embodiments, the pulse widths can include those falling within a range including from at least 10 ns to 200 ns. In some embodiments, the pulse widths can include those falling within a range including from at least 20 ns to 100 ns. In another embodiment, the pulse widths can include those falling within a range including from at least 50 ns to 1500 ns. In other embodiments, the pulse widths can include those falling within a range including from at least 1 ns to 5000 ns. Still alternatively, the pulse widths can fall outside of the foregoing ranges.

[0164] Exemplary nanosecond lasers can include those within the UV to IR spectrum, spanning wavelengths of about 10 nanometers to 1 millimeter. In some embodiments, the light sources suitable for use in the catheter systems herein can include those capable of producing light at wavelengths of from at least 750 nm to 2000 nm. In some embodiments, the light sources can include those capable of producing light at wavelengths of from at least 700 nm to 3000 nm. In some embodiments, the light sources can include those capable of producing light at wavelengths of from at least 100 nm to 10 micrometers (μm). Nanosecond lasers can include those having repetition rates of up to 200 kHz. In some embodiments, the laser can include a Q-switched thulium:yttrium-aluminum-garnet (Tm:YAG) laser. In some embodiments, the laser can include a neodymium:yttrium-aluminum-garnet (Nd:YAG), holmium:yttrium-aluminum-garnet (Ho:YAG), erbium:yttrium-aluminum-garnet (Er:YAG), excimer laser, helium-neon laser, carbon dioxide laser, as well as doped, pulsed, fiber lasers.

Pressure Waves

[0165] The catheters herein can generate pressure waves having maximum pressures in the range of at least 1 mega-

pascal (MPa) to 100 MPa. The maximum pressure generated by a particular catheter will depend on the light source, the absorbing material, the bubble expansion, the propagation medium, the balloon material, and other factors. In some embodiments, the catheters herein can generate pressure waves having maximum pressures in the range of at least 2 MPa to 50 MPa. In other embodiments, the catheters herein can generate pressure waves having maximum pressures in the range of at least 2 MPa to 30 MPa. In yet other embodiments, the catheters herein can generate pressure waves having maximum pressures in the range of at least 15 MPa to 25 MPa. In some embodiments, the catheters herein can generate pressure waves having peak pressures of greater than or equal to 1 MPa, 2 MPa, 3 MPa, 4 MPa, 5 MPa, 6 MPa, 7 MPa, 8 MPa, 9 MPa, 10 MPa, 11 MPa, 12 MPa, 13 MPa, 14 MPa, 15 MPa, 16 MPa, 17 MPa, 18 MPa, 19 MPa, 20 MPa, 21 MPa, 22 MPa, 23 MPa, 24 MPa, or 25 MPa, 26 MPa, 27 MPa, 28 MPa, 29 MPa, 30 MPa, 31 MPa, 32 MPa, 33 MPa, 34 MPa, 35 MPa, 36 MPa, 37 MPa, 38 MPa, 39 MPa, 40 MPa, 41 MPa, 42 MPa, 43 MPa, 44 MPa, 45 MPa, 46 MPa, 47 MPa, 48 MPa, 49 MPa, or 50 MPa. It is appreciated that catheters herein can generate pressure waves having operating pressures or maximum pressures that can fall within a range, wherein any of the foregoing numbers can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

[0166] Therapeutic treatment can act via a fatigue mechanism or a brute force mechanism. For a fatigue mechanism, operating pressures would be about at least 0.5 MPa to 2 MPa, or about 1 MPa. For a brute force mechanism, operating pressures would be about at least 20 MPa to 30 MPa, or about 25 MPa. Pressures between the extreme ends of these two ranges may act upon a calcified lesion using a combination of a fatigue mechanism and a brute force mechanism.

[0167] The pressure waves described herein can be imparted upon the treatment site from a distance within a range from at least 0.1 millimeters (mm) to 25 mm extending radially from a longitudinal axis of a catheter placed at the treatment site. In some embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 10 mm to 20 mm extending radially from a longitudinal axis of a catheter placed at the treatment site. In other embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 1 mm to 10 mm extending radially from a longitudinal axis of a catheter placed at the treatment site. In yet other embodiments, the pressure waves can be imparted upon the treatment site from a distance within a range from at least 1.5 mm to 4 mm extending radially from a longitudinal axis of a catheter placed at the treatment site. In some embodiments, the pressure waves can be imparted upon the treatment site from a range of at least 2 MPa to 30 MPa at a distance from 0.1 mm to 10 mm. In some embodiments, the pressure waves can be imparted upon the treatment site from a range of at least 2 MPa to 25 MPa at a distance from 0.1 mm to 10 mm. In some embodiments, the pressure waves can be imparted upon the treatment site from a distance that can be greater than or equal to 0.1 mm, 0.2 mm, 0.3 mm, 0.4 mm, 0.5 mm, 0.6 mm, 0.7 mm, 0.8 mm, or 0.9 mm, 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or 10 mm, or can be an amount falling within a range between any of the foregoing.

[0168] It should be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

[0169] It should also be noted that, as used in this specification and the appended claims, the phrase “configured” describes a system, apparatus, or other structure that is constructed or configured to perform a particular task or adopt a particular configuration. The phrase “configured” can be used interchangeably with other similar phrases such as arranged and configured, constructed and arranged, constructed, manufactured and arranged, and the like.

[0170] All publications and patent applications in this specification are indicative of the level of ordinary skill in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated by reference.

[0171] As used herein, the recitation of numerical ranges by endpoints shall include all numbers subsumed within that range (e.g., 2 to 8 includes 2.1, 2.8, 5.3, 7, etc.).

[0172] The headings used herein are provided for consistency with suggestions under 37 CFR 1.77 or otherwise to provide organizational cues. These headings shall not be viewed to limit or characterize the invention(s) set out in any claims that may issue from this disclosure. As an example, a description of a technology in the “Background” is not an admission that technology is prior art to any invention(s) in this disclosure. Neither is the “Summary” or “Abstract” to be considered as a characterization of the invention(s) set forth in issued claims.

[0173] The embodiments described herein are not intended to be exhaustive or to limit the invention to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and understand the principles and practices. As such, aspects have been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope herein.

[0174] It is understood that although a number of different embodiments of the catheter systems have been illustrated and described herein, one or more features of any one embodiment can be combined with one or more features of one or more of the other embodiments, provided that such combination satisfies the intent of the present invention.

[0175] While a number of exemplary aspects and embodiments of the catheter systems have been discussed above, those of skill in the art will recognize certain modifications, permutations, additions and sub-combinations thereof. It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations, additions and sub-combinations as are within their true spirit and scope, and no limitations are intended to the details of construction or design herein shown.

What is claimed is:

1. A balloon catheter system comprising:
a catheter configured to advance to a treatment site, the catheter comprising an elongate shaft and a balloon that

is coupled to the elongate shaft, the catheter defining an inflation fluid pathway including an inflation lumen and an interior of the balloon that are in fluid communication;

- a fortified balloon inflation fluid disposed within the inflation fluid pathway, the fortified balloon inflation fluid comprising a base inflation fluid and fortification component including at least one of carbon and iron, the fortified balloon inflation fluid configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid; and
 - a first energy guide that is disposed along the elongate shaft and is positioned at least partially within the balloon, the first energy guide being configured to be in communication with an energy source and including an end disposed within the balloon.
2. The catheter system of claim 1, wherein the fortification component includes an iron dextran.
 3. The catheter system of claim 1, wherein the fortification component includes nanoparticles.
 4. The catheter system of claim 1, wherein the fortification component includes one of iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, gold-coated carbon nanotubes and copper-coated carbon nanotubes.
 5. The catheter system of claim 1, wherein the base inflation fluid includes saline and contrast medium in a ratio of saline to contrast medium of from 25:75 volume percent to 75:25 volume percent.
 6. The catheter system claim 1, wherein the fortification component is configured to modify one of viscosity, density and surface tension of the fortified balloon inflation fluid compared to the base inflation fluid.
 7. A balloon catheter system comprising:
 - a catheter configured to advance to a treatment site, the catheter comprising an elongate shaft and a balloon that is coupled to the elongate shaft, the catheter defining an inflation fluid pathway including an inflation lumen and an interior of the balloon that are in fluid communication;
 - a fortified balloon inflation fluid disposed within the inflation fluid pathway, the fortified balloon inflation fluid comprising a base inflation fluid and fortification component including at least one of carbon and iron, the fortified balloon inflation fluid configured to reduce a threshold for inducing plasma formation and a shockwave in the fortified balloon inflation fluid compared to the base inflation fluid; and
 - a first energy guide that is disposed along the elongate shaft and is positioned at least partially within the balloon, the first energy guide being configured to be in communication with an energy source and including an end disposed within the balloon.
 8. The catheter system of claim 7, wherein the fortification component includes an iron dextran.
 9. The catheter system of claim 7, wherein the fortification component includes nanoparticles.

10. The catheter system of claim 7, wherein the fortification component includes one of iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, gold-coated carbon nanotubes and copper-coated carbon nanotubes.

11. The catheter system of claim 7, wherein the base inflation fluid includes saline and contrast medium in a ratio of saline to contrast medium of from 25:75 volume percent to 75:25 volume percent.

12. The catheter system claim 7, wherein the fortification component is configured to modify one of viscosity, density and surface tension of the fortified balloon inflation fluid compared to the base inflation fluid.

13. A method for generating pressure waves to induce fractures at a treatment site within or adjacent a vessel wall of a blood vessel, the method comprising:

advancing a catheter to the treatment site within the blood vessel, the catheter comprising an elongate shaft, a balloon coupled to the elongate shaft, and an energy guide disposed along the elongate shaft so that the energy guide is positioned at least partially within the balloon;

expanding the balloon with a fortified balloon inflation fluid, the fortified balloon inflation fluid comprising a base inflation fluid and a fortification component, the fortified balloon inflation fluid being configured to reduce a threshold for inducing plasma formation in the fortified balloon inflation fluid compared to the base inflation fluid, wherein the fortification component includes one of carbon and iron; and

activating an energy source in communication with the energy guide and the fortified balloon inflation fluid to provide pulses of energy from the energy source to the fortified balloon inflation fluid so that plasma formation occurs in the fortified balloon inflation fluid.

14. The method of claim 13, wherein the fortification component comprises one of iron dextran and nanoparticles.

15. The method of claim 13, wherein the fortification component comprises nanoparticles, the nanoparticles including one of iron nanoparticles, gold nanoparticles, copper nanoparticles, carbon nanoparticles, carbon nanotubes, gold-coated carbon nanotubes and copper-coated carbon nanotubes.

16. The method of claim 13, wherein the energy guide comprises an optical fiber and wherein the energy source comprises a laser.

17. The method of claim 13, wherein the pulses of energy from the energy source are delivered at a frequency of between 1 Hz and 30 Hz.

18. The method of claim 13, wherein the pulses of energy from the energy source are delivered at a frequency of between 30 Hz and 5000 Hz.

19. The method of claim 13, wherein the base inflation fluid comprises saline and contrast medium in a ratio of saline to contrast medium of 25:75 volume percent to 75:25 volume percent.

20. The method of claim 13, wherein plasma formation creates a shockwave in the fortified balloon inflation fluid.

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